

Intrinsic Heralding and Optimal Decoders for Non-Abelian Topological Order

Dian Jing,^{1,2,*} Pablo Sala,^{3,4} Liang Jiang,² and Ruben Verresen^{2,†}

¹*Department of Physics, University of Chicago, Chicago, IL 60637, USA*

²*Pritzker School of Molecular Engineering, University of Chicago, Chicago, IL 60637, USA*

³*Department of Physics and Institute for Quantum Information and Matter,
California Institute of Technology, Pasadena, CA 91125, USA*

⁴*Walter Burke Institute for Theoretical Physics, California Institute of Technology, Pasadena, CA 91125, USA*

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Topological order (TO) provides a natural platform for storing and manipulating quantum information. However, its stability to noise has only been systematically understood for Abelian TOs. In this work, we exploit the non-deterministic fusion of non-Abelian anyons to inform active error correction and design decoders where the fusion products, instead of flag qubits, herald the noise. This intrinsic heralding enhances thresholds over those of Abelian counterparts when noise is dominated by a single non-Abelian anyon type. Furthermore, we use Bayesian inference to obtain a statistical mechanics model for fixed-point non-Abelian TOs with perfect measurements under any noise model, which yields the optimal threshold conditioned on measuring anyon syndromes. We numerically illustrate these results for $D_4 \cong \mathbb{Z}_4 \rtimes \mathbb{Z}_2$ TO. In particular, for non-Abelian charge noise and perfect syndrome measurement, we find a conditioned optimal threshold $p_c = 0.218(1)$, whereas an intrinsically heralded minimal-weight perfect-matching (MWPM) decoder already gives $p_c = 0.20842(2)$, outperforming standard MWPM with $p_c = 0.15860(1)$. Our work highlights how non-Abelian properties can enhance stability, rather than reduce it, and discusses potential generalizations for achieving fault tolerance.

Introduction. Topological orders (TOs) are long-range entangled topological phases characterized by ground state degeneracy and anyon excitations [1–9]. They have been exploited to encode and manipulate quantum information thanks to their robustness against local noise [10–15]. TOs are either Abelian or non-Abelian. For Abelian TOs, such as the toric code, the error correction problem—which is to find the homology class of the physical error string with a given set of syndromes—has been well studied [14–16].

In contrast, the error correction problem of non-Abelian TOs is more difficult due to non-Abelian braiding statistics and non-deterministic fusion of anyon excitations [17]. Previous work proved the existence of an error correction threshold for non-Abelian TOs with perfect measurements and numerically demonstrated comparable thresholds for both Abelian and non-Abelian TOs using clustering decoders and renormalization group (RG) decoders [18–27] (for a passive error correction approach, see Ref. 28). However, unlike Abelian TOs, no optimal decoder is known for non-Abelian TOs. Moreover, although the existence of a threshold for continuous error correction in the presence of measurement errors has been proven for the special case of acyclic non-Abelian TOs, none of the previously studied decoders has inspired such a proof for the general non-Abelian case [26, 29, 30].

By definition, non-Abelian anyons, denoted by a , have multiple fusion channels

$$a \times \bar{a} = 1 + b + \dots, \quad (1)$$

leading to a quantum dimension $d_a > 1$. In contrast to the Abelian case, this indeterminacy of fusion implies that moving non-Abelian anyons requires a linear-depth circuit [31–34], which cannot be achieved by local error channels. In this work, we utilize the information left behind by imperfect anyon strings [35, 36] to build ‘intrinsically heralded’ decoders, as illustrated in Fig. 1, which operates without the need for flag qubits [28, 37–41]. This uniquely non-Abelian property leads to improved thresholds compared to Abelian counterparts. Furthermore, we present an optimal decoder for non-Abelian TOs given perfect anyon syndromes, and discuss possible extensions to cases with measurement errors.

While much of our discussion is general, we numerically confirm and illustrate our findings for the non-Abelian $D_4 \cong \mathbb{Z}_4 \rtimes \mathbb{Z}_2$ TO, for it has been recently realized in trapped ions [42, 43] and provides a resource for universal quantum computation [30]. Our analytic and numerical analysis is worked out for the kagome lattice model of Ref. 44, with technical details in the Supplemental Material [45]. A key property of interest is that the D_4 TO has four charge anyons: three Abelian charge anyons transforming under the one-dimensional representations $s_{i=1,2,3}$ of the D_4 gauge symmetry, and a non-Abelian anyon transforming under the two-dimensional representation [2], with the fusion $[2] \times [2] = 1 + s_1 + s_2 + s_3$.

Unheralded decoders of non-Abelian TO. Quantum information encoded in the topological degeneracy of a TO is vulnerable to random unitary noise channels that drive the system into a mixed state. The goal of quantum error correction is to restore the system to its original state without corrupting the encoded information. Concretely, this requires the combined error and correction string to have trivial homology [14, 15].

* rossoneri@uchicago.edu

† verresen@uchicago.edu

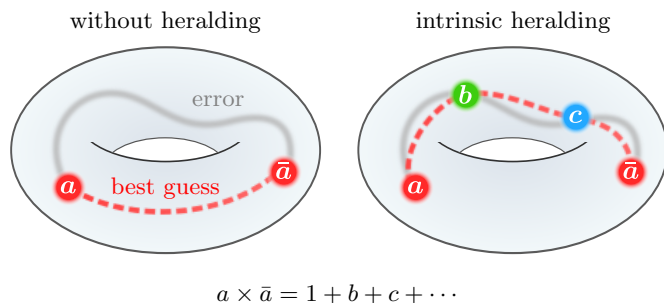


FIG. 1. **Intrinsic heralding from non-Abelian anyons.**

A constant-depth error string not only creates a pair of non-Abelian anyons at its endpoints, but also leaves behind a superposition over possible fusion outcomes along its path. Intermediate anyon syndromes can be extracted by collapsing this superposition, providing information about the original error path. This additional information can improve error correction, particularly when the noise model is biased toward the non-Abelian anyon of interest. Such intrinsic heralding arises from the non-deterministic fusion of non-Abelian anyons, without the need for flag qubits.

To extract information about the physical error, one measures the anyon content of the decohered mixed state, which serves as the error syndrome. When anyon a is not among its own fusion outcomes, that is, it does not appear on the right-hand side of Eq. 1, the a -anyons can be paired with their antiparticles via shortest-length paths, a strategy known as minimal-weight perfect matching (MWPM) decoding [14, 46]. For instance, if the D_4 TO is subjected to incoherent pair-creation of the non-Abelian charge, the MWPM of the [2] anyons exhibits a threshold of $p_c = 0.15860(1)$, as shown by the black square in Fig. 2, coinciding with that of the toric code on a triangular lattice [47, 48]. The threshold is discussed in the End Matter, and details of its numerical determination are provided in the Supplemental Material [45].

When anyon a is among its own fusion outcomes, an odd number of a -anyons may appear following the introduction of pair-creation errors, rendering matching decoders inapplicable. As a result, previous work has relied on multiple iterations of perfect measurement and correction, even for one type of anyon [19–23]. However, these previous threshold estimates have relied exclusively on numerical results, and unlike the Abelian case, the error correction problem has not been mapped to a statistical mechanics (stat-mech) model. Nevertheless, the existence of thresholds under arbitrary local error channels with perfect measurements has been proven using an argument based on error clusters [24–26, 49–51]. If error correction prevents independently created clusters from growing and percolating into larger ones, anyons within each cluster will always fuse to the vacuum. When clusters are much smaller than the code distance (i.e., the size of the logical operator), logical errors are necessarily avoided. The error correction threshold is then defined as the physical error rate below which large clusters, and

hence logical errors, become increasingly rare with increasing code distance [24–26].

Since anyons within a cluster will always fuse to the vacuum, regardless of their internal braiding and fusion, the above arguments apply to any TO. Indeed, previous studies have reported similar thresholds for clustering decoders and Harrington’s RG decoder across Abelian TOs, the Ising anyon code, and the Fibonacci anyon code [18–23, 25–27, 52].

Intrinsically Heralded Decoder for non-Abelian TOs. While the proof and numerical determination of error correction thresholds are valuable, it remains unclear how close these thresholds are to optimal. Moreover, although previously studied decoders have achieved comparable thresholds for Abelian and non-Abelian TOs, they have neither utilized the properties of the TOs to inform decoding nor provided insight into whether the non-Abelian nature offers any advantage for error correction.

A distinctive feature of non-Abelian anyons is that their movement requires a linear-depth quantum circuit. In contrast, physical noise, such as the single-qubit Pauli channel, corresponds to finite-depth operations and cannot generate exact non-Abelian error strings beyond pair-creation on neighboring sites. As a result, along a string of physical error qubits, non-Abelian anyons are created at the endpoints, while superpositions of vacuum and intermediate anyons are also present along the path, reflecting the unresolved non-Abelian fusion channels. The syndromes of both non-Abelian and intermediate anyons can be extracted by measuring the commuting projectors of the Hamiltonian that defines a fixed-point TO

$$H = \sum_i A_i, \quad (2)$$

where the eigenvalues of A_i are 0 or 1 [10, 53]. When the only errors are incoherent pair-creations of non-Abelian a -anyons, the detection of other types of intermediate anyons (e.g., b in Eq. 1) provides a clear signature that the error string has acted on the site. Therefore, we can modify unheralded decoders to require the error correction string to pass through all intermediate anyons. This constitutes the intrinsic heralding that informs and enhances error correction, as shown in Fig. 1.

For the D_4 TO in the presence of incoherent pair-creation of the non-Abelian charge anyons, the intrinsically heralded MWPM decoder selects the shortest correction string that traverses all measured Abelian charges. This achieves a threshold of $p_c = 0.20842(2)$ (yellow symbol in Fig. 2) for logical errors arising from homologically nontrivial [2] anyon loops, a significant increase compared to that of the unheralded MWPM decoder, which has $p_c = 0.15860(1)$ (black square), without needing flag qubits. See Supplemental Material [45] for numerical details.

While the intrinsically heralded decoder improves identification of the physical error string, full recovery of the encoded quantum state still requires multiple rounds of measurement and error correction due to the non-

deterministic fusion of non-Abelian anyons [19–23]. After each round, existing anyons are removed, while newly created anyons along the correction string can be addressed by subsequent applications of the intrinsically heralded decoder. Since minimum-weight decoders do not increase the size of independently created error clusters [24], the clusters arising from error correction remain non-percolating as long as the physical error parameters lie within the error-correcting phase. In other words, the threshold against incoherent pair-creation errors of a single anyon type is set by the first application of intrinsically heralded minimum-weight decoding. Confirming this in practice, we find that when accounting for logical errors arising from both non-Abelian [2] charges as well as the Abelian charges created during the fusion in the correction step, the thresholds for the D_4 TO in the presence of [2] anyon noise are found to be $p_c = 0.1586(2)$ for unheralded MWPM and $p_c = 0.2084(5)$ for intrinsically heralded MWPM, both within one standard deviation of the thresholds considering only nontrivial [2] loops.

In the simulations described above and detailed in the End Matter and the Supplemental Material [45], a single application of finite-depth circuits along the MWPM path corrects the [2] anyons, followed by another for the Abelian anyons. More generally, one round of finite-depth correction suffices to annihilate all a -anyons generated by incoherent pair-creation noise, provided a is not among its own fusion outcomes, with subsequent rounds removing the intermediate anyons introduced during correction. In contrast, when a is among its own fusion outcomes, annihilating a -anyons along the minimal error tree, known as the Steiner minimal tree in computer science [54], requires resources equivalent to perfect measurement and correction using linear-depth circuits repeated for a logarithmic number of iterations in the linear system size. Indeed, in each iteration, the linear-depth circuit fuses anyons within each connected component of the tree, reducing the number of anyons by at least half.

Optimal Decoding for non-Abelian TOs. Instead of correcting along the shortest error string, a decoder may consider all admissible error configurations. To the best of our knowledge, this has thus far been explored only in the context of optimal decoding of Abelian TOs. The seminal work by Dennis *et al.* [14] on the toric code can be interpreted as calculating the probability of homology class h given anyon syndromes $\mathbf{a}$:

$$P(h|\mathbf{a}) = \sum_{E \in h} P(E|\mathbf{a}) \propto \sum_{E \in h, \partial E = \mathbf{a}} \left(\frac{p}{1-p} \right)^{|E|}, \quad (3)$$

where the sum is over error paths E for uncorrelated noise, and two paths are said to be in the same homology class if they can be continuously deformed into one another. The error-correcting phase is characterized by the dominance of a single class in the infinite code distance limit, which determines the optimal decoding strategy. We remark that this approach can also be applied to correcting non-Abelian a -anyons when a is not among

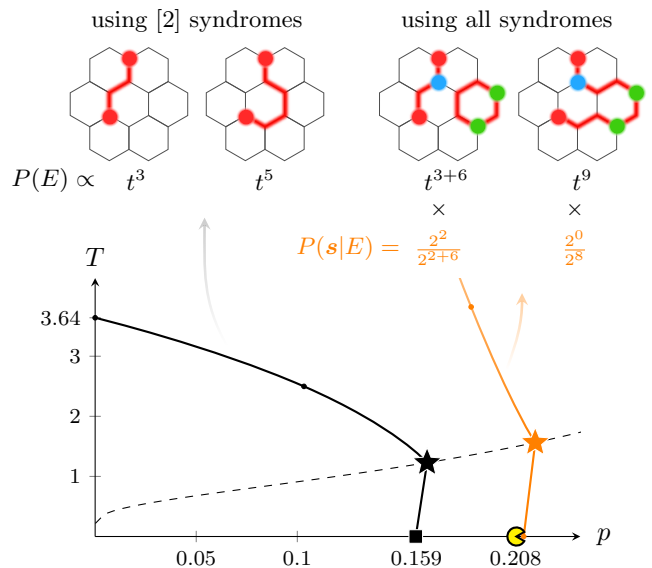


FIG. 2. Error correction thresholds for D_4 TO with charge noise. The phase diagram of the stat-mech models to which we map the error correction problems of the D_4 TO in the presence of non-Abelian charge noise, with example snapshots. The black phase boundary corresponds to the random-bond Ising model on the triangular lattice, associated with the unheralded matching decoder which only sees the non-Abelian charge anyon (red dot in snapshots); the black square and star mark the thresholds of the MWPM and maximum-likelihood decoders (Eq. 3), respectively. The unheralded matching decoder considers only $P(E) \propto t^{|E|}$ with $t = e^{-2\beta}$. The yellow symbol is the intrinsically heralded MWPM decoder, where the error string is forced to pass through the Abelian charge fusion products (blue and green dots in snapshots), with threshold $p_c = 0.20842(2)$. The orange phase boundary corresponds to the threshold using the full set of syndromes, where each snapshot is weighted by $P(\mathbf{s}|E)P(E)$ (Eq. 4). Here, $P(\mathbf{s}|E)$ (Eq. 5) includes a factor of $\frac{1}{2}$ for each vertex of the error string not passing through a non-Abelian anyon, and a factor of 2 for each Abelian parity constraint, as illustrated by the last two error strings above the phase diagram. The optimal decoder is marked by the orange star along the Nishimori line (dashed), $t = \frac{p}{1-p}$, with $p_c = 0.218(1)$. Details on the stat-mech models, the calculation of $P(\mathbf{s}|E)$, and the numerical simulations can be found in the Supplemental Material [45].

its own fusion outcomes. For the D_4 TO under [2] anyon noise, using only the [2] syndromes yields a threshold no better than $p_c = 0.1642(3)$, corresponding to the multicritical point of the random-bond Ising model on the triangular lattice [55–60], shown in Fig. 2 as the black star, with details provided in the End Matter.

To find an optimal decoder for non-Abelian TOs under arbitrary noise, we consider decoding conditioned on the full syndrome $\mathbf{s} = \{\mathbf{a}, \mathbf{b}, \dots\}$. Reliable inference of the physical error E is only possible if the conditional probability $P(E|\mathbf{s})$ (for fixed $\mathbf{s}$) lies in a ‘short string’ phase. Although $P(E|\mathbf{s})$ is a challenging quantity to calculate,

we can make it tractable by applying Bayes' theorem:

$$P(E|\mathbf{s}) = \frac{P(\mathbf{s}|E)P(E)}{P(\mathbf{s})} \propto P(\mathbf{s}|E)P(E). \quad (4)$$

While $P(E) \propto (\frac{p}{1-p})^{|E|}$ represents the probability of physical single-qubit errors occurring along the string E , $P(\mathbf{s}|E)$ accounts for the probabilistic collapse of superpositions over non-Abelian fusion channels into a specific set of intermediate anyons along E . This quantity is given by the expectation value of the anyon projectors evaluated on the corrupted quantum state $|E\rangle$:

$$P(\mathbf{s}|E) = \langle E | \prod_i [(1 - \lambda_i)(1 - A_i) + \lambda_i A_i] | E \rangle, \quad (5)$$

where $\lambda_i \in \{0, 1\}$ is the measurement outcome of projector A_i . The optimal threshold is then identified with a ‘string proliferation’ phase transition of the quenched-disorder stat-mech model, where syndrome $\mathbf{s}$ is the ‘disorder’ variable, and $P(E|\mathbf{s})$ is a stat-mech model on the configuration space of errors. Since expectation values are insensitive to scaling stat-mech weights by an arbitrary function $f(\mathbf{s})$, we can choose analytic weights $P(\mathbf{s} \cap E) = P(\mathbf{s}|E)P(E)$, such that the partition function $Z_{\mathbf{s}} = \sum_E P(\mathbf{s} \cap E)$ equals the quenched disorder probability, $Z_{\mathbf{s}} = P(\mathbf{s})$, known as the Nishimori property [14, 61–68].

If the noise model corresponds to incoherent pair-creation in the anyon basis, the stat-mech model gives an explicit optimal decoder that corrects with the error string in the most likely homology class, which maximizes $P(h|\mathbf{s}) \propto \sum_{E \in h} P(\mathbf{s}|E)P(E)$. More generally, if the noise model creates coherence in the anyon basis, the optimal threshold given by the stat-mech model can be achieved by decomposing the physical error into effective incoherent anyon strings and decoding with anyon strings in the most likely homology class for all anyons. Further details on optimal decoding and the corresponding stat-mech models are provided in the End Matter. Although non-deterministic fusion can create new anyons after the correction step, the initial step determines the true bottleneck and thus sets the threshold p_c , since errors not in the confined phase at this point cannot result in confined downstream fusion products, as previously reasoned for MWPM decoders and supported by a detailed argument in the End Matter.

The key challenge in determining the optimal threshold for non-Abelian TOs lies in efficiently computing $P(h|\mathbf{s})$, which in turn depends on the efficient determination of $P(\mathbf{s}|E)$. In the Supplemental Material [45], we design an efficient Monte Carlo protocol to sample contributing terms in $P(h|\mathbf{s})$ for the D_4 TO [69–72], with example calculations of $P(\mathbf{s}|E)$ illustrated in Fig. 2. For D_4 TO under [2] anyon noise, we find the optimal threshold to be $p_c = 0.218(1)$, corresponding to the orange star at the intersection of the orange phase boundary in Fig. 2 and the Nishimori line $\beta = \ln \sqrt{\frac{1-p}{p}}$ (dashed). This shows

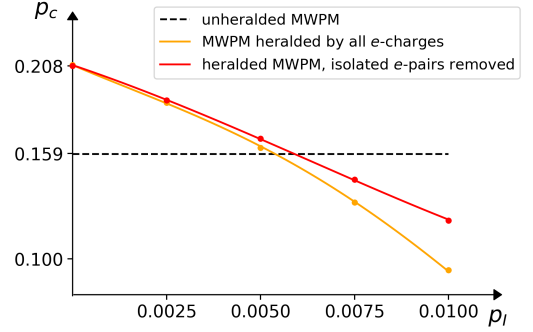


FIG. 3. **Stability of heralding.** Error correction threshold of non-Abelian charges in the D_4 TO as a function of the intermediate Abelian charge pair-creation rate p_I . The yellow solid line shows thresholds from a naïve application of the intrinsically heralded MWPM decoder for non-Abelian charges, while the red solid line reflects thresholds after incorporating an algorithm that identifies isolated Abelian charge pairs. The black dashed line indicates the unheralded MWPM threshold. The advantage from intrinsic heralding persists up to $p_I \approx 0.5\%$ when the decoder is applied naïvely, and is further improved by the algorithm.

that the threshold of the intrinsically heralded MWPM decoder is close to optimal. Furthermore, in the Supplemental Material [45], we give a stat-mech model for the D_4 TO with arbitrary Pauli noise, leaving the question of its efficient sampling to future work.

Stability of Heralding. When intermediate anyons, such as the Abelian charges in the D_4 TO, are pair-created by the error channel, they can lead to incorrect heralding of non-Abelian anyons. Nevertheless, if the noise is strongly biased toward pair-creating non-Abelian a -anyons and intermediate anyon fluctuations are low, intrinsically heralded decoders for the a -anyons can still be naïvely applied as if the intermediate anyon information were reliable. Indeed, as shown by the yellow solid line in Fig. 3, the threshold improvement for non-Abelian charges over the unheralded MWPM decoder (black dashed line) in the D_4 TO persists up to an intermediate anyon (i.e., Abelian charge) pair-creation rate of $p_I \approx 0.5\%$.

The optimal threshold for non-Abelian anyons with perfect measurements, in the presence of intermediate anyon noise, still follows Eq. 4. For the D_4 TO, the corresponding stat-mech model for Eq. 5 is also given in the Supplemental Material [45]. However, unlike the case where noise pair-creates only one type of anyon, the error E now contains strings of multiple anyon types that terminate in more complex ways, complicating efficient sampling. Nonetheless, the naïve heralded MWPM approach can be improved by simple modifications, such as ignoring clusters of intermediate anyons far from non-Abelian anyons. As a proof of principle, we removed isolated pairs of Abelian charges from heralding non-Abelian charges in the D_4 TO [45] and observed an improvement as demonstrated by the red solid line in Fig. 3. This shows that

intrinsically heralded decoders can be modified to remain effective beyond the regime of strongly biased noise.

Measurement Errors. In practice, measurement errors can occur, so anyon syndromes must be continuously measured and physical errors continuously corrected as they arise [14, 20, 25]. Physical and measurement errors are identified by changes in the outcomes of consecutive measurements of the commuting projectors in the non-Abelian TO. Alternatively, suppose that anyon syndromes are extracted by measuring terms in a quasi-stabilizer Hamiltonian whose non-Pauli terms commute only within the ground-state subspace [28, 44]. In that case, frequent fluctuations of intermediate anyons can serve as reliable indicators of measurement errors, effectively serving as a weak time-like heralding. Just as intrinsic heralding improves the performance of MWPM decoders with perfect measurements over their unheralded counterparts, applying intrinsic heralding in both space and time under noisy measurements can similarly aid the identification of the error homology class and enhance error correction performance when combined with correction protocols such as clustering or local cellular automata methods [25, 26, 30].

When anyon a is not among its own fusion outcomes, identifying the pair-creation and measurement error string for a reduces to a matching problem and is expected to exhibit threshold behavior, as it can be mapped onto a three-dimensional (3D) stat-mech model, similar to that in Ref. 14, where the additional dimension represents time. An example of identifying measurement errors and constructing intrinsically heralded matching for the D_4 TO is given in the Supplemental Material [45]. With the addition of intermediate anyon pair-creation and measurement errors, the error E contains multiple types of anyon strings, similar to the case of unstable heralding with perfect measurements, with time-like intermediate anyon strings that may terminate on space-like non-Abelian a -anyon strings.

When anyon a is among its own fusion outcomes, error strings can pass through (in addition to terminating at) a -anyons, and decoding proceeds by constructing a Steiner tree whose anyon terminals lie in two or three dimensions, depending on whether measurements are perfect or noisy, respectively. However, the Steiner tree problem has not been studied in the context of quantum error correction, and previous studies on Fibonacci anyons have avoided this formulation in favor of an iterative, growing-cluster decoder [22, 23]. Nevertheless, the existence of an error correction threshold for general non-Abelian TOs with measurement errors could be proved if the associated Steiner tree were shown to undergo a phase transition from a ‘small tree’ phase to a proliferation phase. The mapping of this transition to that of a classical stat-mech model and its numerical determination remains an open problem for future work.

Conclusion and Outlook. In this work, we exploited the non-deterministic fusion of non-Abelian anyons to enable intrinsically heralded decoding, which significantly

improves the error correction threshold compared to direct application of decoders developed for Abelian TOs. To illustrate this, we numerically determined the thresholds of the D_4 TO with perfect anyon syndromes, where non-Abelian charges may fuse into Abelian ones. Furthermore, we demonstrated that the advantage of intrinsic heralding persists even under unstable heralding.

A natural next step for intrinsic heralding is to study the Steiner tree problem for non-Abelian anyons that are among their own fusion outcomes, with the simplest example being Fibonacci anyons, which intrinsically herald their own error correction. Another example is the S_3 TO, which supports universal quantum computation with information encoded in the internal degrees of freedom of non-Abelian anyons, rather than in the ground-state subspace [73–75].

We also identified the stat-mech model for optimal decoding under arbitrary local noise, for which efficient sampling remains an open problem. More generally, it would be interesting to characterize the phase diagram of these novel stat-mech models, where the quantum dimension of the proliferated anyon enters directly into the Boltzmann weights and thus affects the resulting phase diagram. Additional insight may be drawn from Ref. 68, which follows a similar Bayesian approach. It would also be interesting to investigate our optimal strategy for Abelian TOs with coherent noise [76–82].

Moreover, our result for optimality is conditional on measuring the anyon syndromes, and it would be interesting to explore whether measuring in other bases can improve the threshold—the recent Refs. 35 and 36 suggest a potential $p_c \approx 0.5$ for pure non-Abelian noise, although no decoder has been constructed. Finally, numerical simulations of the 3D matching or Steiner tree problem for continuous error correction with noisy measurements remain to be carried out.

Challenges remain in designing decoders for non-Abelian TOs. A key difficulty in continuous error correction with noisy measurements is that error information can be hidden in the internal degrees of freedom of non-Abelian anyons, making it inaccessible to anyon syndrome measurements. For example, in the D_4 TO, proliferating non-Abelian charges can hide Abelian charges from anyon syndrome measurements, making logical information encoded in the Abelian charges uncorrectable. Therefore, it may be helpful to go beyond the measurements of anyon occupation and even explore adaptive measurement bases that evolve over time. Moreover, incorporating intrinsic heralding into the design of cellular automata decoders, which solely rely on local resources [83], offers another promising direction for future work.

Note added. While this manuscript was being finalized, a preprint appeared which mentioned heralded decoding of non-Abelian S_3 TO, although a specific decoder was not proposed [84].

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- mech model for the D_4 TO with arbitrary Pauli noise in App. B; (3) details of numerical simulations in App. C; and (4) extensions to time-like heralding against measurement errors in App. D.
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End Matter

After physical errors act on the quantum information initially encoded in $\rho_0 = |\psi_0\rangle\langle\psi_0|$, the state is a decohered mixed state $\sum_E P(E) \hat{E} \rho_0 \hat{E}^\dagger$, where $\hat{E}$ denotes a unitary error occurring with probability $P(E)$. Anyon measurements collapse the density matrix, making it

block diagonal in the anyon basis,

$$\rho = \bigoplus_s P(s) \rho_s, \quad (6)$$

where ρ_s is the normalized density matrix associated with

the block labeled by syndromes $\mathbf{s}$. Each block generally has dimension larger than one, arising from the internal degrees of freedom of non-Abelian anyons or from distinct homology classes on topologically nontrivial manifolds. As the internal degrees of freedom are not accessible through measurements, we henceforth assume they have been traced out.

Starting from Eq. 6, we rederive Eq. 4 and verify the validity of the Bayesian inference. On an infinitely large lattice with only trivial homology, ρ is diagonal in the anyon basis after tracing over internal degrees of freedom. As discussed in the main text, the error correction threshold can be identified with the phase transition of a quenched-disorder stat-mech model whose partition function, for each syndrome measurement outcome $\mathbf{s}$ and satisfying the Nishimori property, is given by $Z_{\mathbf{s}} = P(\mathbf{s}) = \text{Tr}(\Pi_{\mathbf{s}} \rho \Pi_{\mathbf{s}})$, where $\Pi_{\mathbf{s}}$ is the projector onto the block labeled by $\mathbf{s}$. Expanding over the configuration space of physical errors yields

$$\begin{aligned} Z_{\mathbf{s}} &= \sum_E P(E) \text{Tr}(\Pi_{\mathbf{s}} \hat{E} \rho_0 \hat{E}^\dagger \Pi_{\mathbf{s}}) \\ &= \sum_E P(E) \langle \psi_0 | \hat{E}^\dagger \Pi_{\mathbf{s}} \hat{E} | \psi_0 \rangle \end{aligned} \quad (7)$$

thus recovering Eq. 4 using Eq. 5 in the main text. Averaging over the ‘disorder’ variable $\mathbf{s}$, the free energy is $\langle \beta F \rangle_{\mathbf{s}} = -\sum_{\mathbf{s}} P(\mathbf{s}) \ln Z_{\mathbf{s}} = -\sum_{\mathbf{s}} P(\mathbf{s}) \ln P(\mathbf{s})$. A singularity in the averaged free energy signals a phase transition of the stat-mech model [14], which, in some cases, can be conveniently detected by sampling order parameters associated with the physical error configurations E .

Quantum information can be encoded using TOs defined on manifolds with nontrivial homology. Within each block $\rho_{\mathbf{s}}$, the states are labeled by the homology classes of all anyon types, $h = \{h_a, h_b, \dots\}$. Here, anyon strings terminating at the same set of syndromes are said to belong to the same homology class if they can be smoothly deformed into one another, or equivalently if the sum of any pair of strings forms only homologically trivial loops. For physical errors E that are *incoherent* in the anyon basis, their homology classes h_E are well-defined, with probabilities given by the partition function

$$Z_{\mathbf{s}, h_E} = P(\mathbf{s}, h_E) = \sum_{E \in h_E} P(E) \text{Tr}(\Pi_{\mathbf{s}} \hat{E} \rho_0 \hat{E}^\dagger \Pi_{\mathbf{s}}). \quad (8)$$

In the error-correcting phase, the probability of a single homology class approaches 1 in the limit of infinite code distance, while those of all other classes vanish. Error correction can then be performed by applying any $\hat{E}$ operator in the dominant class [14].

In general, physical errors are coherent in the anyon basis, leading to ill-defined homology h_E , and therefore cannot be directly used for error correction. They can, however, be decomposed as $\hat{E} = \sum_{\epsilon} \omega(\epsilon|E) \hat{\epsilon}$, where $\hat{\epsilon}$ consists of *incoherent* anyons strings that are consistent with syndromes $\mathbf{s}$ and have support contained within

that of $\hat{E}$. Consequently, the matrix elements of $\rho_{\mathbf{s}}$ are

$$\langle h_L | \rho_{\mathbf{s}} | h_R \rangle = \frac{1}{P(\mathbf{s})} \sum_E \sum_{\substack{\epsilon_L \in h_L \\ \epsilon_R \in h_R}} P(E) \omega(\epsilon_L|E) \omega^*(\epsilon_R|E), \quad (9)$$

where $h_{L,R}$ are the homology classes of all anyon types. While the diagonal entries of $\rho_{\mathbf{s}}$ are real and positive, the off-diagonal elements are generally complex and nonzero.

To perform optimal decoding, one applies anyon strings in the most likely homology class, corresponding to $\arg \max_h \langle h | \rho_{\mathbf{s}} | h \rangle$ [79]. The error-correcting phase of the decoder is defined as the region in the parameter space of the physical errors where, in the limit of infinite code distance, a unique homology class h_i dominates with $\langle h_i | \rho_{\mathbf{s}} | h_i \rangle = 1$, while all other homology classes satisfy $\langle h_j \neq h_i | \rho_{\mathbf{s}} | h_j \neq h_i \rangle = 0$. This definition yields the same phase transition, or equivalently optimal threshold, as the singularity in $\langle \beta F \rangle_{\mathbf{s}}$ on the infinitely large lattice [14, 46]. When a diagonal element of a density matrix is zero, the requirement of positive semi-definiteness forces the entire corresponding row and column to vanish. Consequently, in the error-correcting phase in the infinite code distance limit, $\rho_{\mathbf{s}}$ has a single diagonal element equal to one, with all other entries vanishing, which implies that error correction succeeds with certainty and the state is restored to $\rho_0 = |\psi_0\rangle \langle \psi_0|$.

For the error correction of the D_4 TO defined on a torus under non-Abelian charge noise, the syndromes $\mathbf{s}$ consist of non-Abelian [2] charges and intermediate anyons I . Consequently, the states in $\rho_{\mathbf{s}}$ are labeled by the homology classes of [2] and all intermediate anyons, $\{h_{[2]}, h_I\}$. The physical errors are incoherent among states labeled by $h_{[2]}$ and coherent among states labeled by h_I . As a result, $\rho_{\mathbf{s}}$ is block-diagonal in $h_{[2]}$,

$$\rho_{\mathbf{s}} = \bigoplus_{h_{[2]}} P(h_{[2]}) \rho_{\mathbf{s}, h_{[2]}}, \quad (10)$$

where each normalized block $\rho_{\mathbf{s}, h_{[2]}}$ consists of states labeled by h_I and has nonzero off-diagonal elements.

Furthermore, the error model requires the intermediate anyon strings to lie along the [2] strings in $\hat{\epsilon}_{L,R}$. This implies that the homology class h_E and $h_{[2]}$ are equivalent, and the probabilities $P(\mathbf{s}, h_E)$ in Eq. 8 are the same as $P(h_{[2]})$ in Eq. 10. As a result, below the same threshold, a unique homology class h_E and the corresponding block $\rho_{\mathbf{s}, h_{[2]}}$ dominates with probability one in the limit of infinite code distance. Because intermediate anyon strings are required to lie along the [2] strings, this definite homology class of the [2] component in $\hat{\epsilon}_{L,R}$ ensures that the homology class of intermediate anyons, h_I , is also unique. Consequently, $\rho_{\mathbf{s}, h_{[2]}}$, and hence $\rho_{\mathbf{s}}$ in Eq. 10, take the desired form, containing a single nonzero diagonal element equal to 1. The fact that a unique $h_{[2]}$ ensures a unique h_I indicates that the threshold for intermediate anyons created along [2] strings lies above that of [2] anyons themselves, a conclusion corroborated by our numerical results. Thus, the decoders defined in Eqs. 8

and 10 exhibit the same optimal threshold, set by that of the [2] anyons.

Optimal decoding can then be performed using the diagonal elements of $\rho_{\mathbf{s}}$, computed via Eq. 10, with anyon string operators employed for error correction. However, to avoid the use of linear-depth non-Abelian anyon strings, we first apply the same finite-depth operators appearing in the physical errors of the most likely homology class, identified using either Eq. 8 or 10, to pair-annihilate the [2] charges, while deferring the pairing of intermediate anyons until after the [2] anyons have been corrected. Since the above analysis establishes that the correction of [2] anyons has the same threshold as the optimal decoders in Eqs. 8 and 10, it suffices to account for the additional coherent errors among intermediate anyons that arise along the finite-depth error correction strings used for the pair-annihilation of [2] anyons to prove that our two-step decoding protocol is optimal.

After applying these strings, the density matrix in Eq. 10 becomes $\rho'_{\mathbf{s}} = \bigoplus_{h_o} P(h_o) \rho'_{\mathbf{s}, h_o}$, where h_o denotes the homology class of the closed loops formed by combining the physical error strings with the error correction strings. $P(h_o)$ corresponds one-to-one to $P(h_{[2]})$, since the error correction strings are also incoherent for [2] anyons. The normalized block $\rho'_{\mathbf{s}, h_o}$ contains both the coherence in $\rho_{\mathbf{s}, h_{[2]}}$ and the additional coherence introduced by the error correction strings. Below the threshold of the decoders defined in Eqs. 8 and 10, and in the limit of infinite code distance, the trivial homology class in $h_o = 1$ dominates with $P(h_o = 1) = 1$. By the same reasoning as before, loops in the trivial class contain no homologically nontrivial components. Consequently, the homology class of intermediate anyons is uniquely fixed and $\rho'_{\mathbf{s}, h_o}$ contains a single diagonal element equal to one, with all other entries vanishing. Hence, the error-correcting phase of the decoders defined in Eqs. 8 and 10 guarantees the correction of coherence among intermediate anyons. This concludes our proof that the two-step decoding protocol against non-Abelian [2] charge noise in the D_4 TO is indeed optimal.

Therefore, to numerically determine the optimal threshold of the D_4 TO (orange star in Fig. 2), we sample the order parameter associated with the physical error E in Eq. 7. The physical error string E and the disorder configuration $\mathbf{s}$ is first randomly generated using a disjoint-set union data structure, ensuring that the resulting parity and entanglement structure of I are consistent with those derived from the stabilizer formalism [28, 45]. Then, the summation in Eq. 7 can be rewritten as the partition function of a stat-mech model of classical Ising spins $\{\sigma\}$,

$$\mathcal{Z}_{\mathbf{s}} \propto \sum_{\text{allowed } \{\sigma\}} 2^C e^{-\beta H}, \quad H = - \sum_{\langle i, j \rangle} \eta_{ij}(E) \tilde{w}_{ij}(\mathbf{s}) \sigma_i \sigma_j, \quad (11)$$

$$\tilde{w}_{ij}(\mathbf{s}) = 1 - n_{I|[\mathbf{2}]} K + \frac{2 - n_{[\mathbf{2}]}}{2} \frac{\ln 2}{2\beta},$$

where $\eta_{ij} = -1$ along the error string E and $\eta_{ij} = 1$ elsewhere, $n_{I|[\mathbf{2}]} \in \{0, 1, 2\}$ is the number of intermediate

or [2] anyons connected to the edge, and C is the number of parity constraints on intermediate anyons. The fluctuations of the Ising spins σ generate all physical error strings satisfying $\partial E = [\mathbf{2}]$. These error strings are required to pass through all intermediate anyons, a condition enforced by introducing an arbitrarily large constant K , and to satisfy the set of constraints C , which are verified using a graph search algorithm and assigned zero weight if violated. The moments of the average magnetization of σ are sampled using local-update Monte Carlo simulations weighted by $e^{-\beta H}$ and reweighted by 2^C to account for nonlocal constraints that are computationally costly to evaluate at each step. Binder cumulants are then computed from these moments, after averaging over the disorder variable $\mathbf{s}$, to determine the optimal threshold through finite-size scaling analysis. Further details can be found in the Supplemental Material [45].

Lastly, we discuss how to construct the unheralded optimal decoder that pair-annihilates incoherently created non-Abelian [2] charges without the knowledge of intermediate anyons. The decohered mixed state after introducing physical errors, on an infinitely large lattice with only trivial homology, takes the block diagonal form

$$\rho = \bigoplus_{[\mathbf{2}]} \rho_{[\mathbf{2}]}, \quad (12)$$

where each block $\rho_{[\mathbf{2}]}$ is spanned by states labeled by anyon configurations, $|\mathbf{s}\rangle = \{[\mathbf{2}], \mathbf{I}\}$, and remains unchanged upon the measurement of only [2] anyons. Furthermore, the diagonal elements of ρ correspond to

$$\begin{aligned} \langle [\mathbf{2}], \mathbf{I} | \rho | [\mathbf{2}], \mathbf{I} \rangle &= \sum_E P(E) \text{Tr}(\Pi_{[\mathbf{2}], \mathbf{I}} \hat{E} \rho_0 \hat{E}^\dagger \Pi_{[\mathbf{2}], \mathbf{I}}) \\ &= \sum_E P(E) P([\mathbf{2}], \mathbf{I} | E). \end{aligned} \quad (13)$$

Since the unheralded decoder does not measure or make use of the syndrome information associated with intermediate anyons, the subspace within each $\rho_{[\mathbf{2}]}$ block corresponding to different intermediate anyon configurations $\mathbf{I}$ but fixed non-Abelian syndromes $[\mathbf{2}]$ can be traced out, yielding a reduced diagonal density matrix

$$\tilde{\rho} = \sum_{[\mathbf{2}]} |[\mathbf{2}]\rangle \langle [\mathbf{2}]| \sum_E P(E) \sum_{\mathbf{I}} P([\mathbf{2}], \mathbf{I} | E), \quad (14)$$

the diagonal elements of which correspond to

$$\begin{aligned} P([\mathbf{2}]) &= \sum_E P(E) \sum_{\mathbf{I}} P([\mathbf{2}], \mathbf{I} | E) \\ &= \sum_E P(E) P([\mathbf{2}] | E) = \sum_{\partial E = [\mathbf{2}]} P(E). \end{aligned} \quad (15)$$

Setting $Z_{[\mathbf{2}]} = P([\mathbf{2}])$ recovers the stat-mech model for the optimal toric code decoder [14], identical to that for the optimal unheralded decoding of non-Abelian [2] charges. The unheralded MWPM decoder is recovered by replacing the sum $\sum_{\partial E = [\mathbf{2}]}$ with its largest term.

Appendix A: Stabilizer Formalism for Error Correction in the D_4 Topological Order

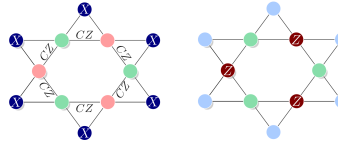
We demonstrate the improved error correction threshold of our intrinsically heralded decoder using the D_4 TO realized in the kagome lattice model of Ref. 44. In this Appendix, we consider a single color of Pauli $\hat{X}$ errors, which create pairs of m -anyons at the endpoints of error strings and leave a superposition of vacuum and e -anyons along their paths. Without loss of generality, we take the Pauli $\hat{X}$ errors to occur on red qubits, resulting in intermediate e -charges on blue and green stars. We analyze the error correction for the D_4 TO under this error channel and derive constraints on the measurement outcomes of intermediate Abelian anyons relevant to our numerical simulations. Additionally, we discuss the calculation of $P(\mathbf{s}|E)$ in Eq. 5 of the main text.

1. D_4 TO on three-colorable kagome lattice

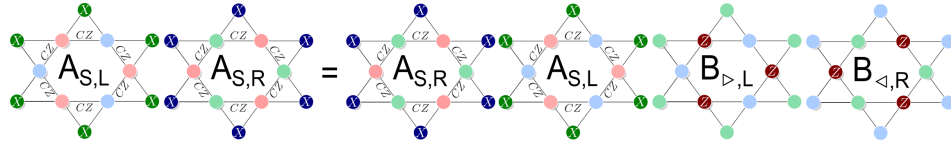
The D_4 TO can be defined on a three-colorable kagome lattice by the quasi-stabilizer Hamiltonian

$$H_{D_4} = - \sum_{s \in \{\star\}} A_s - \sum_{t \in \{\triangleright, \triangleleft\}} B_t, \quad (\text{A1})$$

where the star and triangle operators are defined as


(A2)

Each star and triangle term in the Hamiltonian has eigenvalues ± 1 [42–44]. In this representation, the TO is a $\mathbb{Z}_2^3$ gauge theory ‘twisted’ to become non-Abelian [44]. Using that terminology, the violations of the star operators create Abelian e -charges, whereas excited triangle operators correspond to non-Abelian m -fluxes. The anyons also carry a color label inherited from the lattice. The fusion rules are $m_R \times m_R = 1 + e_B + e_G + e_B e_G$, $e_B \times e_B = 1$, $e_G \times e_G = 1$, and their color permutations. Notably, the non-Abelian anyons fuse to Abelian ones, which is an example of being acyclic [26]. Besides multiple fusion channels for the m -anyons, the non-Abelian nature of the TO is also manifested in the following commutation relation


(A3)

where L and R are adjacent stars. While all Hamiltonian terms commute in the 22-fold degenerate ground space, the presence of an m -anyon causes the star operator whose support overlaps with the anyon to no longer commute with three of its neighboring stars. The measurement of such star operators should therefore be avoided.

Remarkably, this ‘twisted $\mathbb{Z}_2^3$ gauge theory’ realizes the same topological order as the D_4 quantum double [44]. If we use the language of D_4 gauge theory, then the non-Abelian charge anyon corresponds to m_c for an arbitrary choice of color. We choose the convention $[2] = m_R$ going forward. Moreover, the three Abelian charges $s_{i=1,2,3}$ of the D_4 gauge theory then correspond to e_B , e_G and $e_B e_G$. Hence, the above fusion rule agrees with the one claimed in the main text. For a full dictionary between the anyons of D_4 gauge theory and the twisted $\mathbb{Z}_2^3$ gauge theory, we refer to the appendix of Ref. 43. Due to the aforementioned equivalence, we will freely switch between talking about the non-Abelian charge anyon $[2]$ and the non-Abelian flux anyon m_R .

Alternatively, the D_4 TO realized in the kagome lattice model can be defined by a commuting-projector Hamiltonian

$$H_{D_4} = \sum_{s \in \{\star\}} A_s^p + \sum_{t \in \{\triangleright, \triangleleft\}} B_t^p, \quad A_s^p = \frac{1 - A_s}{2} \frac{1 + B_{\triangleright}}{2} \frac{1 + B_{\triangleleft}}{2}, \quad B_t^p = \frac{1 - B_t}{2}. \quad (\text{A4})$$

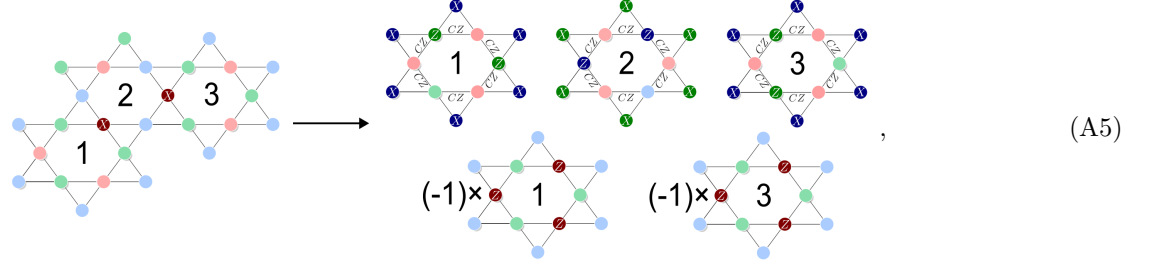
Since all terms in the Hamiltonian commute with one another, they can be simultaneously measured. If an m -anyon is present on a star operator A_s^p , i.e., $B_{\triangleright}^p = 1$ or $B_{\triangleleft}^p = 1$, the measurement of that star yields a value of 0. This commuting-projector Hamiltonian formulation is particularly useful for decoding in the presence of measurement errors, but the operator A_s^p is more difficult to measure experimentally than A_s .

In our numerical simulations, we adopt the quasi-stabilizer Hamiltonian formulation of the D_4 TO.

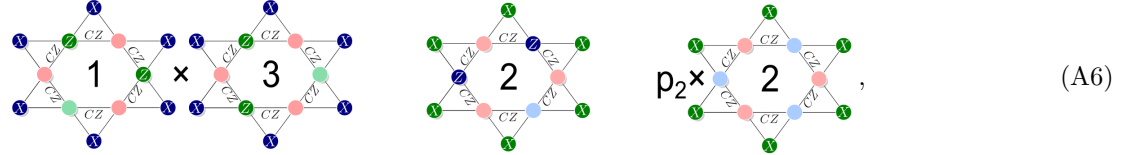
2. Constraints on e -charge measurements following single color of Pauli $\hat{X}$ errors

In this subsection, a simplified version of the stabilizer formalism for the D_4 TO developed in Ref. 28 is used to derive the parity constraints on the intermediate e -charge measurements after the introduction of red Pauli $\hat{X}$ errors. Under this noise channel, blue and green star operators are affected, as well as the red triangle operators located on those stars. The blue and green stars together form a honeycomb lattice, with each color occupying one of its two sublattices.

The ground states of a D_4 TO do not have anyon excitations. Therefore, the stabilizer generators of the ground states on a torus consist of the Hamiltonian terms, namely the star and triangle operators in Eq. A2, along with six logical operators [43, 44]. After the application of a short red Pauli $\hat{X}$ string, the stabilizers are modified as follows:

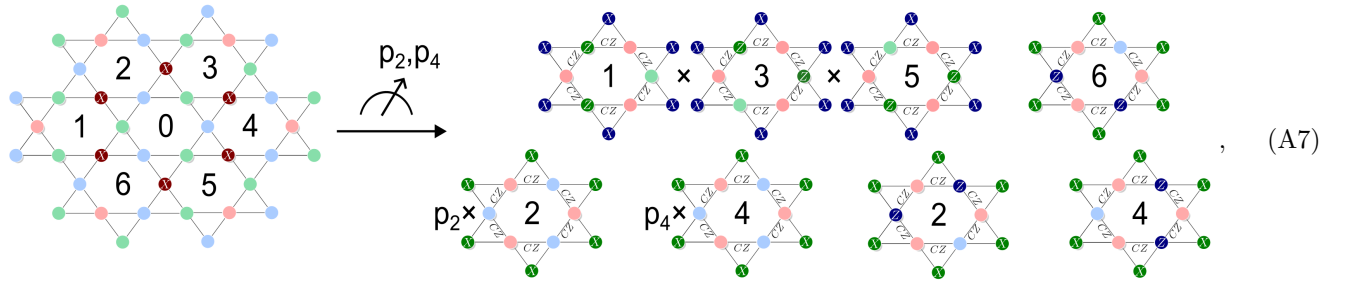


while all other stabilizers are unchanged. Since the Pauli $\hat{X}$ string ends on stars 1 and 3, these two stars will not be measured. The stabilizer $p_2 A_2$ resulting from the measurement of star 2 anti-commutes with the modified star stabilizers 1 and 3. Consequently, the product of the modified star stabilizers 1 and 3, together with $p_2 A_2$, becomes the new stabilizer generators following the measurement, replacing modified star stabilizers 1 and 3:

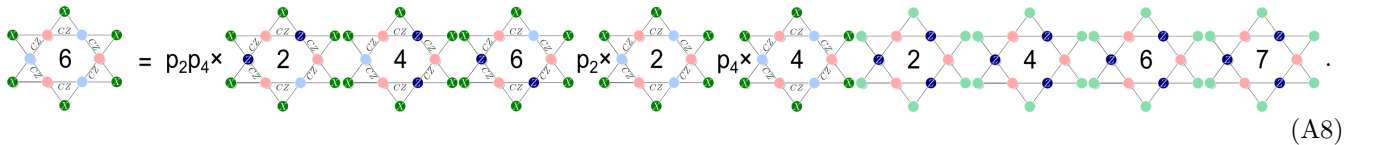


where the measurement outcome, p_2 , is $+1$ (vacuum) or -1 (e -charge) with equal probability $\frac{1}{2}$. The triangle stabilizers commute with the measurement of star 2 and are therefore unaffected.

A parity constraint is relevant when the Pauli $\hat{X}$ string forms a closed loop. Following the analysis above, the star stabilizers are replaced by the following stabilizers after the measurement of stars 2 and 4:

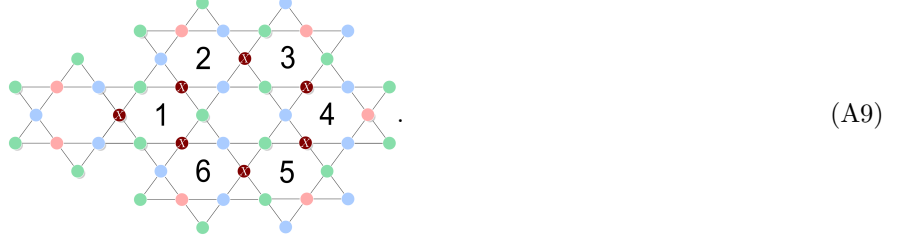


while the triangle stabilizers remain unchanged. Since star 6 commutes with all stabilizers in Eq. A7, it must be a product of stabilizers up to a constant:



This leads to a deterministic outcome $p_6 = p_2 p_4$, reflecting the requirement that the total number of green e -charges on stars 2, 4, and 6 is even. A similar analysis shows that the total number of blue e -anyons on stars 1, 3, and 5 must also be even.

One can also consider the case where the error string forms a branched loop:



Using the stabilizer formalism, one can show that the even-parity constraint on green stars 2, 4, and 6 still holds. However, since star 1 supports an m -anyon and is therefore not measured, the constraint on blue stars 1, 3, and 5 no longer applies, and the measurement outcomes of stars 3 and 5 can take arbitrary values of ± 1 .

Therefore, there is one even-parity constraint for each non-branching, homologically trivial Pauli $\hat{X}$ loop for each color of e -anyons. This is the only constraint on the measurement outcomes after the introduction of Pauli $\hat{X}$ errors.

On a non-branching Pauli $\hat{X}$ loop with nontrivial homology, the constraints on e -anyon measurement depend on the logical sector of the initial state that we are trying to protect. One might suspect that this would lead to different error correction thresholds for different logical sectors. However, an unbranched homologically nontrivial loop is extremely unlikely, and such a loop is not observed in any of the over 10^8 error configurations in our simulation of the matching decoder. Therefore, it was concluded that all ground states of the D_4 TO have the same error correction threshold, and we did not specify the initial state on which error correction was performed in this paper.

3. Value of $P(\mathbf{s}|E)$ in Eq. 5

For the D_4 TO, Eq. 5 in the main text can be rewritten as

$$P(\mathbf{s}|E) = \langle E | \prod_{s \in \{\star\}} [(1 - \lambda_s)(1 - A_s^p) + \lambda_s A_s^p] \prod_{t \in \{\triangleright, \triangleleft\}} [(1 - \lambda_t)(1 - B_t^p) + \lambda_t B_t^p] | E \rangle, \quad (\text{A10})$$

where $\mathbf{s} = \{\mathbf{m}_R, \mathbf{e}_B, \mathbf{e}_G\}$ under red Pauli $\hat{X}$ errors. The state $|E\rangle$ is an eigenstate of the triangle operators, as shown in Eq. A5, with $\lambda_{t \in \partial E} = 1$ at the endpoints of the error string and $\lambda_{t \notin \partial E} = 0$ elsewhere, corresponding to syndromes $\mathbf{m}_R = \partial E$. Furthermore, projectors A_s^p have an eigenvalue $\lambda_s = 0$ at these endpoints, $s \in \partial E$, and away from the error string, $s \notin E$.

Whenever the state is in an eigenstate of a commuting projector A_s^p or B_t^p , the expectation values of $(1 - \lambda_s)(1 - A_s^p) + \lambda_s A_s^p$ or $(1 - \lambda_t)(1 - B_t^p) + \lambda_t B_t^p$ are equal to one. Therefore,

$$P(\mathbf{s}|E) = \langle E | \prod_{s \in \{\wedge\} = E \setminus \partial E} [(1 - \lambda_s)(1 - A_s^p) + \lambda_s A_s^p] | E \rangle, \quad (\text{A11})$$

where $\wedge$ denotes the set of blue and green stars on which Pauli $\hat{X}$ errors have occurred on exactly two red qubits, and the relation $\{\wedge\} = E \setminus \partial E$ holds due to the geometry of the kagome lattice.

On each star $s \in \wedge$, a measurement of A_s^p is performed, producing the results 0 (vacuum) and 1 (e -charge), each with equal probability $\frac{1}{2}$. Equivalently, this corresponds to the operator $(1 - \lambda_s)(1 - A_s^p) + \lambda_s A_s^p$ having expectation value $\frac{1}{2}$ for both $\lambda_s = 0$ and $\lambda_s = 1$. However, when the error string E forms an isolated closed loop of any color, the e -charge measurement outcomes are subject to a parity constraint, as demonstrated in the previous subsection. When the constraint is violated by the syndromes $\mathbf{s}$, the expectation value in Eq. A11 vanishes, indicating that E is not a valid error string to be considered for optimal decoding. When the syndromes satisfy the constraint, the measurement outcome of one star along the isolated closed loop is determined by the outcomes of all other stars on the loop. Consequently, the expectation value of $(1 - \lambda_s)(1 - A_s^p) + \lambda_s A_s^p$ for this particular star is one.

Denoting the total number of parity constraints in E as C , we can determine the value of $P(\mathbf{s}|E)$ by

$$P(\mathbf{s}|E) = 2^{C-\wedge} = 2^{C+\underbrace{\quad}_{\text{Y-shaped intersections}} + \frac{|\mathbf{m}|}{2} - |E|}, \quad (\text{A12})$$

where $\underbrace{\quad}_{\text{Y-shaped intersections}}$ is the number of Y-shaped intersections in E which always host an m -flux, $\mathbf{m}$ is the total number of red m -fluxes in the syndrome $\mathbf{s}$, and $|E|$ is the length of the error string. Eq. A12 is valid as long as all constraints C are satisfied by the syndromes $\mathbf{s}$ and $\mathbf{m}_R = \partial E$; otherwise, $P(\mathbf{s}|E) = 0$, thereby disallowing the error string E .

In our numerical simulations where only red Pauli $\hat{X}$ errors are considered, we use Eq. A12, whereas the corresponding expression for Eq. 5 in the main text under the full Pauli error channel, along with its mapping to a local statistical mechanics model, is provided in Appendix B.

4. Constraints on e -charge measurements after correcting m -fluxes

In this subsection, we derive the parity constraints on e -charge measurements following the correction of m -fluxes using red Pauli $\hat{X}$ strings, and discuss their implications for determining logical errors.

After the measurements yielding the stabilizers in Eq. A6, error correction can be performed by applying Pauli $\hat{X}$ operators to the same qubits as the error qubits in Eq. A5, namely the red qubits between star 1 and star 2, and between star 2 and star 3. By removing the m -fluxes, this operation restores the quantum state to the $+1$ eigenstate of all triangle operators, leading to new stabilizers:

(A13)

All other stabilizers remain unchanged. In the subsequent round of e -charge measurements, all star operators commute and can be measured simultaneously, as there are no remaining m -fluxes. In particular, the first stabilizer in Eq. A13 imposes an even parity constraint on stars 1 and 3, resulting in the same measurement outcome p for both. Furthermore, the stabilizer resulting from the measurement of either star 1 or 3 anti-commutes with only the third stabilizer in Eq. A13, and therefore replaces it, resulting in updated stabilizers:

(A14)

For each connected component in the union of the physical error and correction Pauli $\hat{X}$ strings, two parity constraints on the e -charge measurements after flux correction arise, one for the blue stars and one for the green stars. This occurs because the stabilizer resulting from an e -charge measurement, whether post-error or post-flux correction, anti-commutes with neighboring star stabilizers linked by the physical error or correction Pauli $\hat{X}$ string, resulting in a new stabilizer given by the product of neighboring stars. The resulting constraints reflect the cumulative effect of two rounds of e -charge measurements. Specifically, using the stabilizer formalism described above, it can be shown that, after flux correction, certain sets of star operators become entangled and are governed by the same constraint, according to the following rules:

Physical error string	Pauli $\hat{X}$ string for flux correction	Entanglement of star operators after flux correction
No error		Entangle neighboring stars to the top and bottom right
		Entangle neighboring stars to the top and bottom right
		Entangle neighboring stars to the top and bottom right
, followed by e measurement	No correction	Entangle neighboring stars to the top and bottom right
, followed by e measurement		Entangle neighboring stars to the top and bottom right*
, followed by e measurement		Entangle all three neighboring stars
		Entangle neighboring stars to the top and bottom right

These rules apply to each blue or green star operator and its green or blue neighbors, respectively. The case marked with * corresponds to the one discussed above.

When the connected component in the union of the physical error and correction Pauli $\hat{X}$ strings is homologically trivial, the constraints on both blue and green stars are even parity constraints. In contrast, when the component is homologically nontrivial, the constraints depend on the logical sector of the initial quantum state. Such homologically nontrivial components often signal the occurrence of logical errors.

For example, if the initial state is stabilized by the horizontal blue logical $-\bar{Z}$ operator and there exists a single homologically nontrivial horizontal loop formed by the red Pauli $\hat{X}$ physical error and correction strings, then the parity constraint on blue stars along this loop is odd, while that on green stars remains even. The same holds for initial states stabilized by green or vertical logical $-\bar{Z}$ operators. Such a nontrivial loop results in a logical $\bar{X}$ error, which is also manifested in the odd number of e -charges remaining after flux correction. In contrast, if the state is stabilized by the vertical blue logical $\bar{X}$ operator and the red Pauli $\hat{X}$ error and correction strings form one or more homologically nontrivial horizontal loops, then no parity constraint is imposed on the blue stars. However,

e -measurements along any such loop project the system into a definite parity sector of blue e -charges, corresponding to a state stabilized by a blue logical $\pm \bar{Z}$ operator, thereby inducing a different type of logical error.

Therefore, we declare a logical error whenever the union of the physical error and correction Pauli $\hat{X}$ strings contains any homologically nontrivial component. If no such component is present, we proceed with correcting the remaining e -charges.

Since no homologically nontrivial component exists in the union of the physical error and correction Pauli $\hat{X}$ strings, there is an even number of e -charges of each color. The quantum state after the second round of e -measurements is thus equivalent to one generated by any configuration of Pauli $\hat{Z}$ strings that pairwise connect same-color charges along the union, as all such configurations are homologically equivalent. Since the e -charges are Abelian, logical $\bar{Z}$ errors arising from homologically nontrivial e -charge loops can be identified in the same way as in the toric code. We declare a logical error whenever the symmetric difference between the effective Pauli $\hat{Z}$ error strings and the e -charge correction strings forms a noncontractible loop on the torus defined by the periodic boundaries.

Appendix B: Statistical Mechanics Model for the D_4 TO against Single-Qubit Pauli Noise

1. Pauli $\hat{X}$ noise on red qubits and Pauli $\hat{Z}$ noise on blue and green qubits

Let us consider the scenario of having X noise on the red qubits of the kagome lattice with error rate p_R^x , and Z noise on both blue and green qubits with error rates p_G^z and p_B^z , respectively. The anyon syndromes, denoted by their respective anyon labels, are obtained from measurements of the commuting projectors defined in Eq. A4. We begin by labeling the presence (absence) of an m_R flux on triangle t by $m_t = 1(0)$, corresponding to $B_t = -1(1)$ or equivalently $B_t^p = 1(0)$, as defined in Eqs. A1, A2, and A4. The centers of these triangles lie on the vertices of the honeycomb lattice introduced in Appendix A and represent the mobility of m_R flux excitations (see upper panels in Fig. 2). Similarly, we denote the presence or absence of Abelian charges by $e = e_s^G, e_s^B$, with each variable taking values 1 or 0. Our goal in this subsection is to compute the probability of a given error string $E = \{E_X^R, E_Z^G, E_Z^B\}$ conditioned on a set of anyon syndromes, and to recast the result as a local statistical mechanics model.

To begin, we use that

$$\text{prob}(E|m_t, e) = \text{prob}(m_t, e|E) \text{prob}(E) \frac{1}{\text{prob}(e, m_t)}. \quad (\text{B1})$$

Hence, we need to compute $\text{prob}(m_t, e|E)$, since

$$\text{prob}(E) \propto \left(\underbrace{\frac{p_R^x}{1 - p_R^x}}_{=t_X^R} \right)^{E_X^R} \left(\underbrace{\frac{p_G^z}{1 - p_G^z}}_{=t_Z^G} \right)^{E_Z^G} \left(\underbrace{\frac{p_B^z}{1 - p_B^z}}_{=t_Z^B} \right)^{E_Z^B}. \quad (\text{B2})$$

To obtain an explicit stat-mech model for the conditional probability, we express $\text{prob}(m_t, e|E)$ as the expectation value of a product of commuting projectors, as given in Eq. 5. To detect the presence or absence of a non-Abelian anyon m_R , we insert the projector $\frac{1}{2}(1 + (1 - 2m_t)B_t)$ at the center of each red triangle, which is equivalent to $(1 - m_t)(1 - B_t^p) + m_t B_t^p$. To detect Abelian e anyons, we place the projector $(1 - e_s)(1 - A_s^p) + e_s A_s^p$ at the center of each star s . This reduces to $\frac{1}{2}(1 + (1 - 2e_s)A_s)$ for stars with $s \notin \partial E_X^R$, and to $1 - e_s$ for stars with $s \in \partial E_X^R$ (and hence vanishes if an m_R and a e anyon lie on the same star, namely if $e \in \partial E_X^R$). All together we find

$$\text{prob}(m_t, e|E) = \langle E | \prod_t \frac{1}{2}(1 + (1 - 2m_t)B_t) \prod_{s \notin \partial E_X^R} \frac{1}{2}(1 + (1 - 2e_s)A_s) | E \rangle, \quad (\text{B3})$$

where $|E\rangle = \prod_{i \in E_X^R} X_i \prod_{j \in E_Z^G \cup E_Z^B} Z_j |D_4\rangle$, and the condition $s \notin \partial E_X^R$, can be explicitly imposed by multiplying by $\prod_s (1 - \delta_{e_s, \partial E_X^R})$. To simplify the notation, let us define $\tilde{m}_t = 1 - 2m_t \in \{-1, +1\}$ and similarly $\tilde{e}_s = 1 - 2e_s$. Equation B3 then takes the form

$$\text{prob}(m_t, e|E) = \langle E | \prod_t \frac{1}{2}(1 + \tilde{m}_t B_t) \prod_{s \notin \partial E_X^R} \frac{1}{2}(1 + \tilde{e}_s A_s) | E \rangle. \quad (\text{B4})$$

As a first step, we can simplify the expression by noticing that

$$\begin{aligned}
\text{prob}(\mathbf{m}_t, e|E) &= \langle D_4 | \prod_{i \in E_X^R} X_i \prod_{j \in E_Z^G \cup E_Z^B} Z_j \prod_t \frac{1}{2}(1 + \tilde{m}_t B_t) \prod_{s \notin \partial E_X^R} \frac{1}{2}(1 + \tilde{e}_s A_s) \prod_{i \in E_X^R} X_i \prod_{j \in E_Z^G \cup E_Z^B} Z_j | D_4 \rangle \\
&= \langle D_4 | \prod_{t \in \partial E_X^R} \frac{1}{2}(1 - \tilde{m}_t B_t) \prod_{t \notin \partial E_X^R} \frac{1}{2}(1 + \tilde{m}_t B_t) \prod_{i \in E_X^R} X_i \prod_{\substack{s \notin \partial E_X^R \\ s \in \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2}(1 - \tilde{e}_s A_s) \prod_{\substack{s \notin \partial E_X^R \\ s \notin \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2}(1 + \tilde{e}_s A_s) \prod_{i \in E_X^R} X_i | D_4 \rangle \\
&= \prod_{t \in \partial E_X^R} \frac{1}{2}(1 - \tilde{m}_t) \prod_{t \notin \partial E_X^R} \frac{1}{2}(1 + \tilde{m}_t) \langle D_4 | \prod_{i \in E_X^R} X_i \prod_{\substack{s \notin \partial E_X^R \\ s \in \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2}(1 - \tilde{e}_s A_s) \prod_{\substack{s \notin \partial E_X^R \\ s \notin \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2}(1 + \tilde{e}_s A_s) \prod_{i \in E_X^R} X_i | D_4 \rangle \quad (\text{B5})
\end{aligned}$$

where in the last equation we used the fact that $B_t |D_4\rangle = |D_4\rangle$. Here, we explicitly keep the first factors (rather than a Kronecker delta), hoping that we can more easily generalize these expressions to the case of imperfect measurements. We now conjugately apply $\prod_{i \in E_X^R} X_i$, using the fact that $X_r C Z_{rj} Z_r = C Z_{rj} Z_j$. Note that the error string E_X^R either does not cross a given star operator A_s , since only stars with $s \notin \partial E_X^R$ contribute to the product, or acts with $\hat{X}$ on two qubits of the same color within the support of A_s . Focusing on the latter, we then find

$$\begin{aligned}
\text{prob}(\mathbf{m}_t, e|E) &= \prod_{t \in \partial E_X^R} \frac{1}{2}(1 - \tilde{m}_t)(1 - \delta_{e_s(t), \partial E_X^R}) \prod_{t \notin \partial E_X^R} \frac{1}{2}(1 + \tilde{m}_t) \prod_{\substack{s \notin \partial E_X^R \\ s \in \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2}(1 - \tilde{e}_s) \prod_{\substack{s \notin \partial E_X^R \\ s \notin \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2}(1 + \tilde{e}_s) \\
&\times \langle D_4 | \prod_{\substack{s \in E_X^R \setminus \partial E_X^R \\ s \in \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2} \left(1 - \tilde{e}_s \left(\prod_{r \in s \cup E_X^R} Z_{r+} Z_{r-} \right) \right) \prod_{\substack{s \in E_X^R \setminus \partial E_X^R \\ s \notin \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2} \left(1 + \tilde{e}_s \left(\prod_{r \in s \cup E_X^R} Z_{r+} Z_{r-} \right) \right) | D_4 \rangle \quad (\text{B6})
\end{aligned}$$

where $e_s(t)$ corresponds to the star s on which the triangle t lies, and $r^\pm$ corresponds to the nearest-neighbor sites of a site r on the central hexagon of the star operator A_s (see Fig. 4). The final step is to perform the ungauging map to the $\mathbb{Z}_2^3$ SPT, as introduced in Ref. 44 (additional details can be found in Ref. 36), followed by a CCZ disentangling circuit, which maps the topological ground state $|D_4\rangle$ to the trivial product state paramagnet $|+\rangle$ and the product of the two Z operators dressing the star operator, to $\tilde{Z}_{r+} \tilde{Z}_{r-}$, where $\tilde{Z}_i$ lies on the sites of the honeycomb lattice where the error string E_X^R lies (see Fig. 6). Overall, we find

$$\begin{aligned}
\text{prob}(\mathbf{m}_t, e|E) &= \underbrace{\prod_{t \in \partial E_X^R} m_t (1 - \delta_{e_s(t), \partial E_X^R}) \prod_{t \notin \partial E_X^R} (1 - m_t)}_{=\prod_t \delta_{m_t, \partial E_X^R} (1 - \delta_{e_s(t), \partial E_X^R})} \underbrace{\prod_{\substack{s \notin \partial E_X^R \\ s \in \partial E_Z^G \cup \partial E_Z^B}} e_s \prod_{\substack{s \notin \partial E_X^R \\ s \notin \partial E_Z^G \cup \partial E_Z^B}} (1 - e_s)}_{=\prod_{s \notin \partial E_X^R} \delta_{e_s, \partial E_Z^G \cup \partial E_Z^B}} \quad (\text{B7}) \\
&\times \langle + | \prod_{\substack{s \in E_X^R \setminus \partial E_X^R \\ s \in \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2} \left(1 - \tilde{e}_s \left(\prod_{r \in s \cup E_X^R} \tilde{Z}_{r+} \tilde{Z}_{r-} \right) \right) \prod_{\substack{s \in E_X^R \setminus \partial E_X^R \\ s \notin \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2} \left(1 + \tilde{e}_s \left(\prod_{r \in s \cup E_X^R} \tilde{Z}_{r+} \tilde{Z}_{r-} \right) \right) | + \rangle, \quad (\text{B8})
\end{aligned}$$

which can be equivalently rewritten as a positive Boltzmann weight (at zero temperature), using one Ising variable per site

$$\begin{aligned}
\text{prob}(\mathbf{m}_t, e|E) &= \prod_t \delta_{m_t, \partial E_X^R} (1 - \delta_{e_s(t), \partial E_X^R}) \prod_{s \notin \partial E_X^R} \delta_{e_s, \partial E_Z^G \cup \partial E_Z^B} \\
&\times \frac{1}{2^{|R|}} \sum_{\{\sigma\}} \prod_{\substack{s \in E_X^R \setminus \partial E_X^R \\ s \in \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2} \left(1 - \tilde{e}_s \left(\prod_{r \in s \cup E_X^R} \sigma_{r+} \sigma_{r-} \right) \right) \prod_{\substack{s \in E_X^R \setminus \partial E_X^R \\ s \notin \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2} \left(1 + \tilde{e}_s \left(\prod_{r \in s \cup E_X^R} \sigma_{r+} \sigma_{r-} \right) \right), \quad (\text{B9})
\end{aligned}$$

where $|R|$ denotes the total number of red qubits after applying the ungauging map.

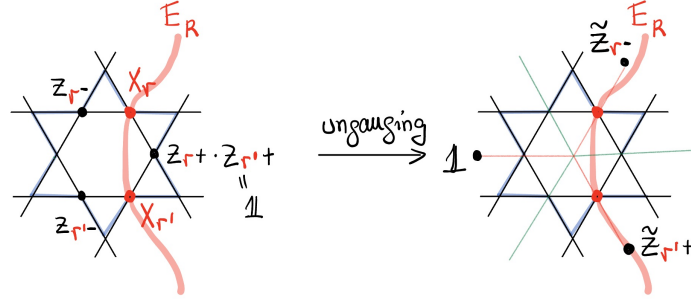


FIG. 4. **Ising strings emerging along E_X^R errors and ungauging.** The conjugate action of $\prod_{i \in E_X^R} X_i$ on star operators A_s^c with $c = G, B$ (as obtained in Eq. B6) dresses A_s^c with a product of two Z 's. The ungauging then maps these two $\tilde{Z}$'s lying along the error string E_X^R .

Putting all together with Eq. B1, and using that $\text{prob}(E|\{m_t, e\})\text{prob}(\{m_t, e\}) = \text{prob}(E \cap \{m_t, e\})$ we then find

$$\begin{aligned} \text{prob}(E \cap \{m_t, e\}) &= (t_X^R)^{E_X^R} (t_Z^G)^{E_Z^G} (t_Z^B)^{E_Z^B} \prod_t \delta_{m_t, \partial E_X^R} (1 - \delta_{e_s(t), \partial E_X^R}) \prod_{s \notin E_X^R} \delta_{e_s, \partial E_Z^G \cup \partial E_Z^B} \\ &\times \frac{1}{2^{|R|}} \sum_{\{\sigma\}} \prod_{\substack{s \in E_X^R \setminus \partial E_X^R \\ s \in \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2} \left(1 - \tilde{e}_s \left(\prod_{r \in s \cup E_X^R} \sigma_r + \sigma_{r-} \right) \right) \prod_{\substack{s \in E_X^R \setminus \partial E_X^R \\ s \notin \partial E_Z^G \cup \partial E_Z^B}} \frac{1}{2} \left(1 + \tilde{e}_s \left(\prod_{r \in s \cup E_X^R} \sigma_r + \sigma_{r-} \right) \right). \end{aligned} \quad (\text{B10})$$

From this expression alone, we can directly conclude the following observations:

- Non-Abelian m_R anyons ($m_t = 1$) need to lie at the boundaries ∂E_X^R .
- Abelian E_Z^G, E_Z^B anyons ($e_s = 1$) need to lie: i) either at the boundaries of $E_Z^G \cup E_Z^B$ but not at E_X^R ; or ii) $s \in E_X^R$.
- If we post-select on all $m_t = 0$ for all t (namely $\partial E_X^R = \emptyset$ and hence all error strings are close loops L_R), then assuming no E_Z^G, E_Z^B errors, we find that

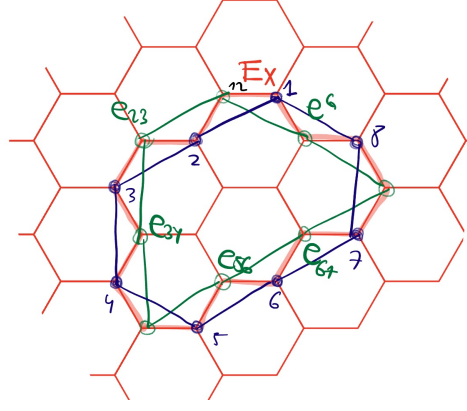
$$\text{prob}(L_R | m_t = 0 \forall t, e) = (t_X^R)^{L_R} \frac{1}{2^{|R|}} \sum_{\{\sigma\}} \prod_{s \in E_X^R} \frac{1}{2} \left(1 + \tilde{e}_s \left(\prod_{r \in s \cup E_X^R} \sigma_r + \sigma_{r-} \right) \right) \prod_{s \in E_X^R} \frac{1}{2} \left(1 + \tilde{e}_s \left(\prod_{r \in s \cup E_X^R} \sigma_r + \sigma_{r-} \right) \right),$$

which for close loops L_R takes the value $\left(\frac{t_X^R}{2}\right)^{L_R} 4^{C_{L_R}}$, where C_{L_R} is the number of connected components in L_R , provided that the parity of anyons along each component is even, i.e., $\prod_{s \in \ell_R} \tilde{e}_s = \prod_{s \in \ell_R} \tilde{e}_s = +1$. Such a configuration is represented in Fig. 5.

2. Full Pauli noise on all qubits

The previous calculation can be extended to a generic single-site Pauli noise model (with both X and Z Pauli channels acting on all sites). Let us denote by e_s^c, m_t^c the charge and flux syndromes associated to a particular color $c = R, G$ or B ; and analogously denote by E_X^c, E_Z^c , error strings created by X or Z Pauli noise respectively, when acting on qubits of color c . We will denote by m_t, e_s the sets of all flux and charge syndromes, respectively, and by E_X, E_Z the sets of all Pauli errors, respectively, across all colors (an example of an allowed error configuration is shown in Fig. 6). Then the conditional probability $\text{prob}(\{m_t^c, e_s^c\} | \{E_Z^c, E_X^c\})$ is given by

$$\begin{aligned} \text{prob}(\{m_t^c, e_s^c\} | \{E_Z^c, E_X^c\}) &= \langle D_4 | \prod_{j \in E_X} X_j \prod_{i \in E_Z} Z_i \prod_t \frac{1}{2} (1 + \tilde{m}_t B_t) \prod_{s \notin \partial E_X} \frac{1}{2} (1 + \tilde{e}_s A_s) \prod_{i \in E_Z} Z_i \prod_{j \in E_X} X_j | D_4 \rangle \\ &= \prod_c \delta_{\partial E_X^c, m_t^c} \prod_{c' \neq c} (1 - \delta_{e_s^{c'}(t), \partial E_X^{c'}}) \langle D_4 | \prod_{j \in E_X} X_j \prod_{\substack{s \notin \partial E_X \\ s \in \partial E_Z}} \frac{1}{2} (1 - \tilde{e}_s A_s) \prod_{\substack{s \notin \partial E_X \\ s \notin \partial E_Z}} \frac{1}{2} (1 + \tilde{e}_s A_s) \prod_{j \in E_X} X_j | D_4 \rangle. \end{aligned} \quad (\text{B11})$$

FIG. 5. $O(4)$ loop model from absence of syndrome fluxes.

Following the same reasoning as in the previous subsection, and denoting by c, c', c'' pairwise unequal colors, we find that

$$\begin{aligned} \text{prob}(\{m_t^c, e_s^c\} | \{E_Z^c, E_X^c\}) &= \prod_{c=R,G,B} \left(\prod_t \delta_{\partial E_X^c, m_t^c} \prod_{c' \neq c} (1 - \delta_{e_s^{c'}(t), \partial E_x^c}) \prod_{s \notin E_x^{c'} \cup E_X^{c''}} \delta_{e_s^c, \partial E_Z^c} \right) \\ &\times \langle D_4 | \prod_{c=R,G,B} \prod_{\substack{s \in E_X^{c'} \cup E_X^{c''} \\ s \notin \partial E_X^{c'} \cup \partial E_X^{c''}}} \frac{1}{2} \left(1 + (-1)^{\delta_{e_s^c, \partial E_Z^c}} e_s^c \prod_{j_{c'} \in s \cap E_X^{c'}} Z_{j_{c'}}^+ Z_{j_{c'}}^- \prod_{j_{c''} \in s \cap E_X^{c''}} Z_{j_{c''}}^+ Z_{j_{c''}}^- \right) | D_4 \rangle, \end{aligned} \quad (\text{B12})$$

where $j_c^\pm$ again correspond to the nearest-neighbor sites of a site j_c of color c , lying on the central hexagon of the star operator as for the derivation in Eq. B6. Hence, $j_c^\pm$ lie on qubits of a different color. Notice that unlike in Eq. B6, now a given star operator A_s^c can be dressed by the nontrivial action of error strings $E_X^{c'}, E_X^{c''}$ corresponding to two different colors, when crossing through A_s^c . Hence, as a result, one finds the product of four (rather than two) Z operators dressing A_s^c . Finally, we further simplify this expression by mapping $(|D_4\rangle \rightarrow |+\rangle)$ via the ungauging map to the $\mathbb{Z}_2^3$ SPT, which maps the dressing Z operators to the product of four $\tilde{Z}$ operators: two lying on the vertices along the $E_X^{c'}$ error string, and the other two along $E_X^{c''}$. Applying the disentangling unitary CCZ, one finds the following expression

$$\begin{aligned} \text{prob}(\{m_t^c, e_s^c\} | \{E_Z^c, E_X^c\}) &= \prod_{c=R,G,B} \left(\prod_t \delta_{\partial E_X^c, m_t^c} \prod_{c' \neq c} (1 - \delta_{e_s^{c'}(t), \partial E_x^c}) \prod_{s \notin E_x^{c'} \cup E_X^{c''}} \delta_{e_s^c, \partial E_Z^c} \right) \\ &\times \langle + | \prod_{c=R,G,B} \prod_{\substack{s \in E_X^{c'} \cup E_X^{c''} \\ s \notin \partial E_X^{c'} \cup \partial E_X^{c''}}} \frac{1}{2} \left(1 + (-1)^{\delta_{e_s^c, \partial E_Z^c}} e_s^c \prod_{j_{c'} \in s \cap E_X^{c'}} \tilde{Z}_{j_{c'}}^+ \tilde{Z}_{j_{c'}}^- \prod_{j_{c''} \in s \cap E_X^{c''}} \tilde{Z}_{j_{c''}}^+ \tilde{Z}_{j_{c''}}^- \right) | + \rangle, \end{aligned} \quad (\text{B13})$$

where $\tilde{Z}_i$ are defined analogously to the previous section (see Fig. 4) but now lying on the triangular lattice displayed in Fig. 6. Alternatively, we can introduce an Ising variable $\sigma = \pm 1$ per vertex on this lattice, and obtain

$$\begin{aligned} \text{prob}(\{m_t^c, e_s^c\} | \{E_Z^c, E_X^c\}) &= \prod_{c=R,G,B} \left(\prod_t \delta_{\partial E_X^c, m_t^c} \prod_{c' \neq c} (1 - \delta_{e_s^{c'}(t), \partial E_x^c}) \prod_{s \notin E_x^{c'} \cup E_X^{c''}} \delta_{e_s^c, \partial E_Z^c} \right) \\ &\times \frac{1}{2^N} \sum_{\{\sigma\}} \prod_{c=R,G,B} \prod_{\substack{s \in E_X^{c'} \cup E_X^{c''} \\ s \notin \partial E_X^{c'} \cup \partial E_X^{c''}}} \frac{1}{2} \left(1 + (-1)^{\delta_{e_s^c, \partial E_Z^c}} e_s^c \prod_{j_{c'} \in s \cap E_X^{c'}} \sigma_{j_{c'}}^+ \sigma_{j_{c'}}^- \prod_{j_{c''} \in s \cap E_X^{c''}} \sigma_{j_{c''}}^+ \sigma_{j_{c''}}^- \right), \end{aligned} \quad (\text{B14})$$

where N is the number of vertices in the resulting triangular lattice. From this expression one can then obtain $\text{prob}(\{E_Z^c, E_X^c\} | \{m_t^c, e_s^c\})$ using Bayes' theorem.

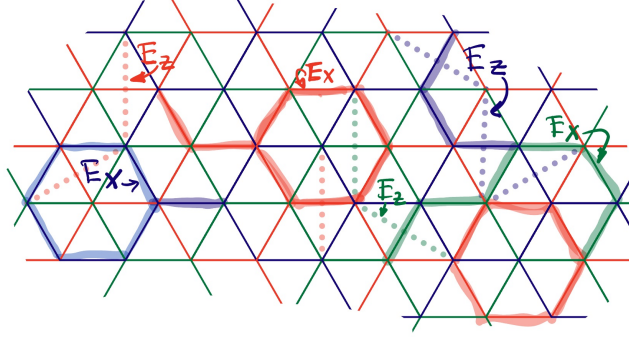
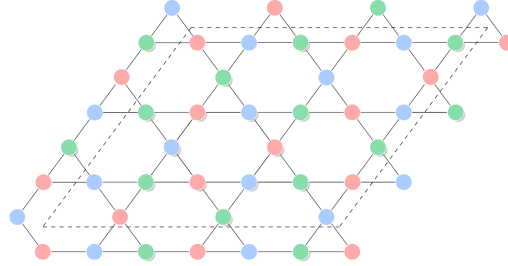


FIG. 6. Error configuration for general noise model.

Appendix C: Numerical Simulation Details

In this subsection, we detail the numerical simulations used to determine the error correction threshold of the D_4 TO under Pauli $\hat{X}$ noise on red qubits and Pauli $\hat{Z}$ noise on blue and green qubits. The D_4 topological order is defined on an $L \times L$ lattice, where L denotes the number of unit cells along each direction, and each unit cell contains 9 stars, as indicated by the boxed region:



(C1)

As discussed later, we introduce classical Ising spins at the centers of the hexagons of the honeycomb lattice, whose vertices correspond to blue and green stars, as described in Appendix A and shown in Fig. 5, or equivalently, at the vertices of the dual triangular lattice, in order to locally deform error strings.

1. Statistical mechanics models

In the first three subsections, we consider red Pauli $\hat{X}$ noise, which pair-creates non-Abelian red m -fluxes, $\mathbf{m}_{red}$, at the endpoints of error strings E and leaves a superposition of vacuum and e -charges along their paths.

Following the seminal work by Dennis *et al.* in Ref. 14, the error correction problem using the unheralded decoders for the m -fluxes is typically analyzed by mapping it to a classical random-bond Ising model (RBIM) defined on the dual triangular lattice:

$$H_1(\{\sigma\}) = - \sum_{\langle i,j \rangle} \eta_{ij}(E) \sigma_i \sigma_j, \quad \mathcal{Z}_{\mathbf{m}_R} = \sum_{\{\sigma\}} e^{-\beta H_1}, \quad (C2)$$

where $\eta_{ij} = -1$ along the error string E and $\eta_{ij} = 1$ elsewhere, with fluctuations in the Ising spins $\{\sigma\}$ generating all homologically equivalent strings satisfying $\partial E = \mathbf{m}_{red}$. The error correction threshold of the unheralded decoders corresponds to the phase transition of the quenched-disorder stat-mech model obtained after summing over the disorder variable $\mathbf{m}_{red}$. In particular, the zero-temperature phase transition corresponds to the unheralded MWPM decoder, while the phase transition along the Nishimori line $\beta = \ln \sqrt{\frac{1-p}{p}}$ corresponds to the maximum likelihood decoder, where p is the rate of red Pauli $\hat{X}$ errors. If the decoder incorrectly overestimates the error rate as $3p$, the threshold is instead given by the phase transition along $\beta = \ln \sqrt{\frac{1-3p}{3p}}$.

With intrinsically heralded decoding, the error correction string is required to pass through all measured e -charges,

$\mathbf{e}_{blue}$ and $\mathbf{e}_{green}$. This problem can be mapped to the following classical local spin model:

$$H_2(\{\sigma\}) = - \sum_{\langle i,j \rangle} \eta_{ij}(E) w_{ij}(\mathbf{e}_B, \mathbf{e}_G) \sigma_i \sigma_j, \quad w_{ij}(\mathbf{e}_B, \mathbf{e}_G) = 1 - n_e K, \quad \mathcal{Z}_s = \sum_{\{\sigma\}} e^{-\beta H_2}, \quad (\text{C3})$$

where $n_e \in \{0, 1, 2\}$ is the number of e -charges connected to the edge, and the value of $\eta_{ij}(E)$ remains the same as above. Here, K is a very large positive number that penalizes error strings bypassing any e -charge. In our simulations, it is set to $27L^2$, which is three times the total number of edges in the honeycomb lattice. As before, the phase transitions of the quenched-disorder stat-mech model obtained after summing over the disorder variable $\mathbf{s}$ correspond to the error correction thresholds of this decoder.

To construct the optimal decoder for the D_4 TO under red Pauli $\hat{X}$ noise, one must account for the conditional probability $P(\mathbf{s}|E)$ in Eqs. 5 and A12, which leads to

$$\mathcal{Z}_s = \sum_{\text{allowed } \{\sigma\}} 2^{C-\wedge} e^{-\beta H_2}, \quad (\text{C4})$$

where the summation runs over all error strings that satisfy $\partial E = \mathbf{m}_{red}$ and the parity constraints on e -charges. Although the number of constraints C is nonlocal, the factor $2^\wedge$ is local and can be absorbed into the definition of the Hamiltonian, leading to:

$$H_3(\{\sigma\}) = - \sum_{\langle i,j \rangle} \eta_{ij}(E) \tilde{w}_{ij}(\mathbf{s}) \sigma_i \sigma_j, \quad \tilde{w}_{ij}(\mathbf{s}) = 1 - n_e K + \frac{2 - n_m \ln 2}{2} \frac{1}{2\beta}, \quad \mathcal{Z}_s = \sum_{\text{allowed } \{\sigma\}} 2^C e^{-\beta H_3}, \quad (\text{C5})$$

where $n_m \in \{0, 1, 2\}$ is the number of m -fluxes connected to the edge. The choice of $\frac{\ln 2}{2\beta}$ is used to implement the factor $2^\wedge$ such that the partition function in Eq. C5 is identical to that in Eq. C4 at all temperatures. The phase transitions of the quenched-disorder stat-mech model away from the Nishimori line correspond to a decoder that incorrectly estimates the error probability p but correctly accounts for the factor $2^{C-\wedge}$. This is meaningful because the factor arises from the fusion channels of the non-Abelian anyons, which are typically known.

We note that in the $p \rightarrow 0$ limit, the Hamiltonian H_3 in Eq. C5 reduces to that of the clean Ising model; however, the 2^C factor remains. Since the clean Ising model contains only closed domain walls (as there are no anyons), and each closed loop satisfies exactly two independent constraints (one for each color), the 2^C factor contributes a multiplicative factor of 4 per domain wall. In the $p \rightarrow 0$ limit, the model reduces to an $O(4)$ loop model on the honeycomb lattice, which is dual to the triangular lattice on which the Ising spins reside. Since this is known to always be in the short-loop phase [86] (at least for the physically accessible parameter regime), the corresponding model in Eq. C5 remains in the ordered phase. We thus conclude that the critical temperature T_c diverges as $p \rightarrow 0$.

2. MWPM simulation details

We first studied the error correction threshold of the MWPM decoders for correcting red non-Abelian m -fluxes arising from red Pauli $\hat{X}$ noise. These thresholds correspond to the zero-temperature phase transition of the stat-mech models in Eqs. C2, C3, or C5. For each MWPM decoder, N_E random error configurations were generated on the edges of an $L \times L$ honeycomb lattice with qubit error rate p , which was varied in increments of 0.001 (or 0.1%) around the threshold, with at least four values sampled both above and below it. If necessary, e -charge measurements were simulated on star operators that do not host any m -flux, with the constraints discussed in Appendix A enforced using a disjoint-set union data structure and a depth-first search algorithm. The matching of m -fluxes was then performed on the honeycomb lattice using the PyMatching Python package [47]. The edge weights on the lattice were set to 1 for the unheralded MWPM decoder, to $w_{ij}(\mathbf{e}_B, \mathbf{e}_G)$ for the heralded MWPM decoder in Eq. C3, and to $\tilde{w}_{ij}(\mathbf{s})$ for the zero-temperature phase transition of the stat-mech model in Eq. C5, where the factor of 2^C was ignored in the simulation of the latter, since the MWPM algorithm disfavors isolated closed loops that give rise to the constraints. A logical error was declared if the union of the physical error string and the flux correction string contained any homologically nontrivial component, as determined by a breadth-first search algorithm on the universal cover. Following the finite-size scaling analysis in Ref. 46, the error correction threshold p_c and the critical exponent ν were determined by fitting the logical error rates P_{logical} for each decoder to

$$P_{\text{logical}} = f(x), \quad x = (p - p_c)L^{1/\nu}. \quad (\text{C6})$$

We also studied the threshold while accounting for logical errors arising from both non-Abelian flux correction and Abelian charge correction. The introduction of physical errors, syndrome measurements, correction using red Pauli $\hat{X}$

strings, and the identification of logical errors due to flux correction were carried out as described above. If no logical error is declared, the simulation proceeds by identifying the blue and green e -charges that arise after flux correction using a disjoint-set union data structure, along with the corresponding effective blue and green Pauli $\hat{Z}$ error strings, following the method outlined in Appendix A. Another round of MWPM decoding was used to determine the shortest Pauli $\hat{Z}$ error correction string, and a logical error was declared if the symmetric difference between the effective Pauli $\hat{Z}$ error strings and the e -charge correction strings contained any noncontractible loop. No logical error is recorded if none is declared after either the flux correction or the charge correction. The threshold p_c and the critical exponent ν were similarly extracted by fitting to Eq. C6.

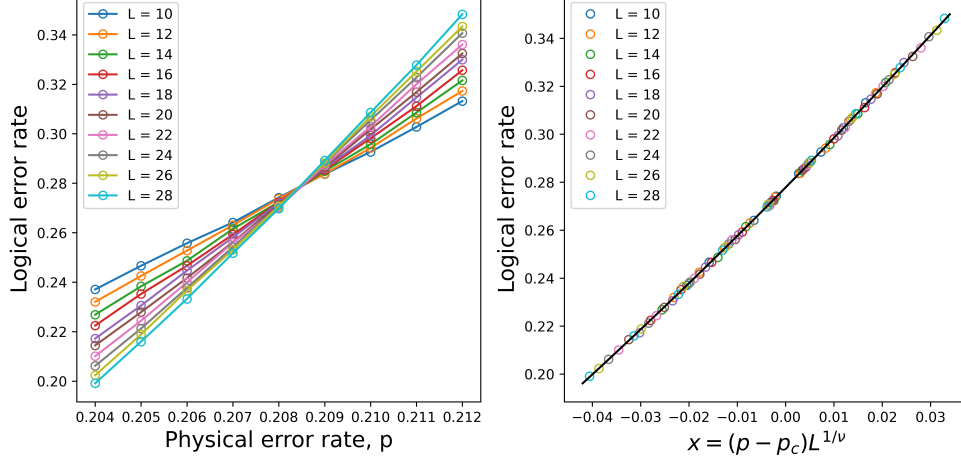
When fitting the numerical data to Eq. C6, the function $f(x) = A + Bx + Cx^2 + EL^{-1/\mu}$ was first used, following the approach in Ref. 46. This model is considered valid if the fitted value of μ differs from zero by at least one standard deviation, which is the case when considering both flux and charge correction steps. Otherwise, the distance-independent function $f(x) = A + Bx + Cx^2$ or $f(x) = A + Bx + Cx^2 + Dx^3$ was used. For the MWPM decoders, both the quadratic and cubic fits yielded similar residual sums of squares, indicating that the quadratic fit $f(x) = A + Bx + Cx^2$ was sufficient.

The parameters and results of the MWPM simulations are summarized in the table below:

Data point	L	N_E	p_c	ν	μ	Fitting model
Unheralded MWPM	10-30 even	10^6	0.15860(1)	1.496(8)		$A + Bx + Cx^2$
Heralded MWPM	10-28 even	10^6	0.20842(2)	1.503(10)		$A + Bx + Cx^2$
Zero-temperature transition of C5	10-28 even	10^6	0.21196(2)	1.488(8)		$A + Bx + Cx^2$
Unheralded MWPM with e -correction	10-26 even	10^6	0.15861(20)	1.552(11)	0.66(46)	$A + Bx + Cx^2 + EL^{-1/\mu}$
Heralded MWPM with e -correction	10-26 even	10^6	0.20844(49)	1.604(21)	0.83(53)	$A + Bx + Cx^2 + EL^{-1/\mu}$

The critical exponents ν listed in this table are all consistent with literature values, approximately 1.5(1), for the RBIM below the Nishimori line [46, 48].

The logical error rates and the finite-size scaling fit for the heralded MWPM decoder with flux correction only, corresponding to the yellow symbol in Fig. 2 of the main text, are shown below as an example demonstrating the validity of the fitting model.



3. Monte Carlo simulation details

When all error strings are considered by the decoder, the thresholds for correcting red non-Abelian m -fluxes correspond to the finite-temperature phase transitions of the stat-mech models in Eqs. C2, C3, and C5, which were numerically determined using Monte Carlo methods.

For each simulation, N_E red Pauli $\hat{X}$ error configurations were generated on the $L \times L$ honeycomb lattice, and the e -charges were measured as described in the previous subsection. For each error and syndrome configuration, the edge weights of the Hamiltonians in Eqs. C2, C3, and C5 were determined accordingly. N_{eq} Metropolis sweeps were performed to equilibrate the system, followed by N_{MC} Monte Carlo sweeps for data collection. In each Metropolis sweep, $3L^2$ Ising spins σ , the total number of spins on the lattice, were randomly selected one at a time and flipped with probability $\min\{e^{-\beta \delta \epsilon}, 1\}$, where $\delta \epsilon$ is the change in energy resulting from the proposed spin flip.

For simulations of the stat-mech models in Eqs. C2 and C3, the magnetization $m = \sum_i \sigma_i$, along with m^2 , m^4 , and m^8 , was calculated after each Monte Carlo sweep. After averaging over all N_{MC} Monte Carlo sweeps and all N_E error and syndrome configurations, the Binder cumulant

$$B = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2} \quad (C7)$$

and its standard deviation were computed from the expectation values $\langle m^2 \rangle$, $\langle m^4 \rangle$, and $\langle m^8 \rangle$.

For simulations of the stat-mech model in Eq. C5, the calculation of the nonlocal factor 2^C is significantly more computationally expensive than performing local spin flips, making it impractical to update C dynamically during each spin flip. To address this issue, the factor 2^C is computed only at the end of each Monte Carlo sweep using a disjoint-set union data structure and a depth-first search algorithm, alongside the calculation of the magnetization m , m^2 , m^4 , m^8 . A constraint check is also performed via depth-first search, and the factor 2^C is set to zero whenever any constraint is violated, thereby excluding disallowed error strings from contributing to the sampling of the partition function. This approach effectively reweights [87–89] the order parameters according to

$$\langle \mathcal{O} \rangle_R = \frac{\langle \mathcal{O} \times 2^C \rangle}{\langle 2^C \rangle}, \quad \mathcal{O} = m, m^2, m^4, m^8, \quad (C8)$$

where $\langle \cdot \rangle$ denotes the average over all samples weighted by $e^{-\beta H_3}$, and $\langle \cdot \rangle_R$ denotes the reweighted average. This reweighting did not compromise the validity of our Monte Carlo sampling, as isolated error loops that give rise to the constraints are rare, and the factor 2^C is nearly uniformly distributed. Indeed, in the representative case of C5 along $\beta = \ln \sqrt{\frac{1-p}{p}}$, corresponding to the optimal decoder indicated by the orange star in Fig. 2 of the main text, with $L = 6$ and $p = 0.218$, the magnetization is $\langle m \rangle = 0.78846 \pm 0.18397$ before weighting and $\langle m \rangle_R = 0.78868 \pm 0.18387$, showing negligible difference, while $\langle 2^C \rangle = 1.004 \pm 0.142$, indicating that the distribution is fairly sharply peaked around unity. The Binder cumulant is then computed using the reweighted expectation values.

The error correction thresholds p_c and the critical exponents ν were determined by fitting the Binder cumulant for each decoder to

$$B = f(x), \quad x = (p - p_c)L^{1/\nu}. \quad (C9)$$

The fitting model $f(x) = A + Bx + Cx^2 + EL^{-1/\mu}$ showed no statistical dependence on L , as the fitted value of μ remained within one standard deviation of zero. Therefore, the quadratic fit $f(x) = A + Bx + Cx^2$ was used for simulations along $\beta = \ln \sqrt{\frac{1-p}{p}}$ (the Nishimori line), $\beta = \ln \sqrt{\frac{2-p}{p}}$, and $\beta = \ln \sqrt{\frac{1-2p}{2p}}$, where it was found to be sufficient, and the cubic fit $f(x) = A + Bx + Cx^2 + Dx^3$ was used for simulations along $\beta = \ln \sqrt{\frac{1-3p}{3p}}$.

The parameters and results of the Monte Carlo simulations are summarized in the table below:

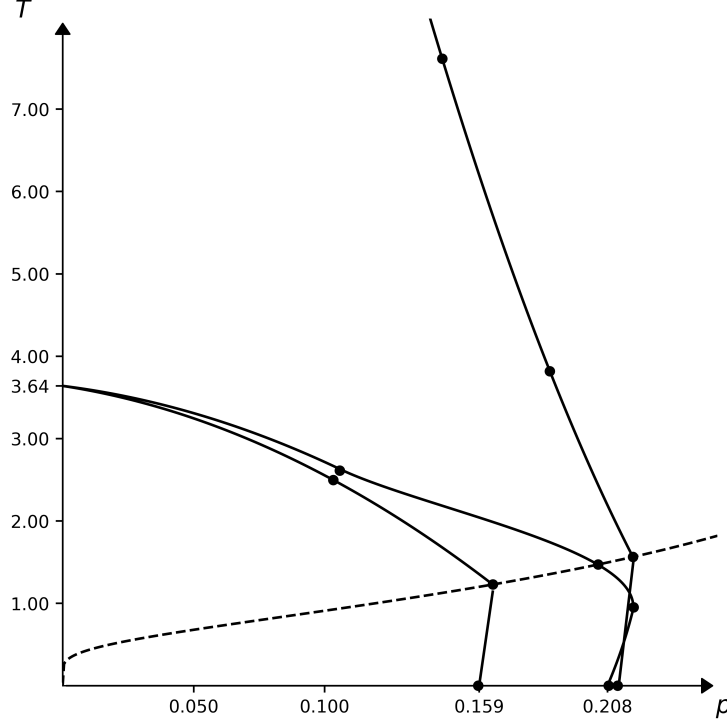
Data point	L	N_E	N_{eq}	N_{MC}	p_c	ν	Fitting model
C2 along $\beta = \ln \sqrt{\frac{1-p}{p}}$	3-9	10^5	3×10^5	3×10^5	0.1642(3)	1.55(20)	$A + Bx + Cx^2$
C2 along $\beta = \ln \sqrt{\frac{1-3p}{3p}}$	3-6	10^5	3×10^5	3×10^5	0.1033(1)	1.19(6)	$A + Bx + Cx^2 + Dx^3$
C3 along $\beta = \ln \sqrt{\frac{1-p}{p}}$	3-6	10^5	5×10^5	3×10^5	0.2044(6)	1.44(32)	$A + Bx + Cx^2$
C3 along $\beta = \ln \sqrt{\frac{1-3p}{3p}}$	3-6	10^5	3×10^5	3×10^5	0.1058(1)	1.16(7)	$A + Bx + Cx^2 + Dx^3$
C3 along $\beta = \ln \sqrt{\frac{2-p}{p}}$	3-6	10^5	5×10^5	3×10^5	0.2180(10)	1.57(60)	$A + Bx + Cx^2$
C5 along $\beta = \ln \sqrt{\frac{1-p}{p}}$	3-6	2×10^5	5×10^5	3×10^5	0.2177(7)	1.70(46)	$A + Bx + Cx^2$
C5 along $\beta = \ln \sqrt{\frac{1-3p}{3p}}$	3-6	2×10^5	5×10^5	3×10^5	0.1449(2)	1.25(9)	$A + Bx + Cx^2 + Dx^3$
C5 along $\beta = \ln \sqrt{\frac{1-2p}{2p}}$	3-6	2×10^5	5×10^5	3×10^5	0.1860(3)	1.33(15)	$A + Bx + Cx^2$

Simulations were not performed for the RBIM in Eq. C2 below the Nishimori line, as its phase diagram is already well established in the literature [55–60].

Still, at or below the Nishimori line, the fitted values of ν for C2 and C3 are consistent with literature values of approximately 1.5(1) for the RBIM [57, 59, 60]. Above the Nishimori line, the phase transitions of C2 and C3 are expected to fall within the Ising universality class, where $\nu = 1$. While our fitted critical exponents suggest that the universality classes above and below the Nishimori line are indeed distinct, the values above the line deviate from

$\nu = 1$, likely due to the limited range of lattice sizes L . For the stat-mech model in Eq. C5, the universality class at phase transition is not known in literature due to the $2^{C-\wedge}$ factor and need not agree with that of the RBIM.

Simulations of the stat-mech model in Eq. C5 below the Nishimori line suffered from poor statistics and fit quality due to long equilibration times. This insufficient equilibration tends to overestimate the transition value of p , and the poor fits provided no indication that the transition below the Nishimori line could exceed the optimal threshold on the line. Therefore, we plot the phase boundary below the Nishimori line as a straight line in the full phase diagram:



where the top line corresponds to the phase boundary of the statmodel in Eq. C5, the middle line to that in Eq. C3, the bottom line to that of the RBIM in Eq. C2, and the dashed line is the Nishimori line $\beta = \ln \sqrt{\frac{1-p}{p}}$. Notably, the maximum p_c along the middle phase boundary does not lie on the Nishimori line. This is because the decoders based on Eq. C3 do not account for the probability of e -charge measurements and overestimate the likelihood of error string deformation using $\frac{p}{1-p}$ per additional weight, whereas the correct value is $\frac{1}{2} \frac{p}{1-p}$.

4. Decoders against Pauli $\hat{X}$ noise on red qubits and Pauli $\hat{Z}$ noise on blue and green qubits

Lastly, we consider decoding in the presence of Pauli $\hat{Z}$ noise on blue and green qubits, which pair-creates blue and green e -charges and leads to incorrect heralding of the red non-Abelian m -fluxes. In this section, we focus exclusively on MWPM decoders, and the reported thresholds on the pair-creation error rate of m -fluxes correspond specifically to the correction of red m -fluxes.

In the physical process, Pauli $\hat{X}$ and Pauli $\hat{Z}$ noise were introduced prior to the measurement of m - and e -anyon syndromes. This is equivalent to first applying red Pauli $\hat{X}$ errors, then measuring the m - and e -syndromes, and finally updating the e -syndromes to account for the effects of Pauli $\hat{Z}$ errors. Therefore, in each simulation, N_E red Pauli $\hat{X}$ error configurations were generated on the $L \times L$ honeycomb lattice, and the e -charges were measured as described previously, before blue and green Pauli $\hat{Z}$ errors were introduced at rate p_z , pair-creating blue or green e -charges on the end of the Pauli $\hat{Z}$ error strings. If a Pauli $\hat{Z}$ error string terminates at a star hosting an m -flux, the syndrome on that star remains unchanged due to the fusion rule $m_R \times e_{B/G} = m_R$.

As described in the main text, decoding can be performed using an intrinsically heralded MWPM decoder, simulated by performing MWPM with edge weights given by $w_{ij}(e_B, e_G) = 1 - n_e K$, where all e -charges e_B and e_G are used to herald the red Pauli $\hat{X}$ error correction string. Logical errors are identified by the presence of homologically nontrivial components in the union of the red Pauli $\hat{X}$ physical error string, and the corresponding logical error rates are fitted according to Eq. C6. The parameters and results of the simulations in which all e -charges are considered are shown as the yellow solid line in Fig. 3 of the main text and are listed in the top four rows of the table below.

Data point	L	N_E	p_c	ν	Fitting model
$pz = 0$	10-28 even	10^6	0.20842(2)	1.503(10)	$A + Bx + Cx^2$
$pz = 0.0025$	10-24 even	10^6	0.18740(3)	1.481(13)	$A + Bx + Cx^2$
$pz = 0.005$	10-24 even	10^6	0.16221(3)	1.501(17)	$A + Bx + Cx^2$
$pz = 0.0075$	10-24 even	10^6	0.13162(4)	1.457(21)	$A + Bx + Cx^2$
$pz = 0.01$	10-24 even	10^6	0.09378(7)	1.522(29)	$A + Bx + Cx^2$
$pz = 0$, isolated e -pairs removed	10-24 even	10^6	0.20796(2)	1.502(13)	$A + Bx + Cx^2$
$pz = 0.0025$, isolated e -pairs removed	10-24 even	10^6	0.18872(3)	1.491(13)	$A + Bx + Cx^2$
$pz = 0.005$, isolated e -pairs removed	10-24 even	10^6	0.16734(3)	1.492(16)	$A + Bx + Cx^2$
$pz = 0.0075$, isolated e -pairs removed	10-24 even	10^6	0.14434(3)	1.511(18)	$A + Bx + Cx^2$
$pz = 0.01$, isolated e -pairs removed	10-24 even	10^6	0.12147(4)	1.476(21)	$A + Bx + Cx^2$

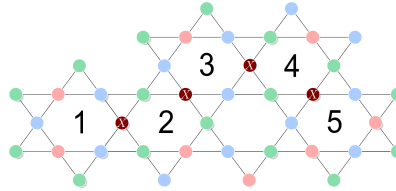
The bottom four rows of the table above record the parameters and results of the simulation in which isolated pairs of e -charges were excluded from heralding, and the corresponding results are plotted as the red solid line in Fig. 3 of the main text. After introducing the Pauli $\hat{Z}$ errors and updating the e -syndromes accordingly, an algorithm was applied to exclude from heralding those pairs of e -charges that are of the same color, are nearest neighbors on the honeycomb sublattice, and have no m -flux on any adjacent star of a different color, in the red Pauli $\hat{X}$ error correction process. The heralded MWPM decoding, identification of logical errors, and the fitting procedure to extract thresholds and critical exponents proceed as described previously.

Appendix D: Measurement Errors

We provide an example illustrating how measurement errors in quasi-stabilizers can be identified from fluctuations of intermediate anyons. This process is trivial for noisy commuting projector measurements. We also discuss the calculation of $P(\mathbf{s}|E)$ in Eq. 5 of the main text in the context of continuous error correction, as a step toward constructing an optimal decoder given noisy anyon syndromes.

1. Example of time-like heralding with measurement errors on non-Abelian flux quasi-stabilizers

A state within the ground state subspace of the D_4 TO, where quantum information is encoded, is prepared at $t = 0$ with no anyon content. At a later time during the continuous error correction process, $t = t_0$, Pauli $\hat{X}$ errors happen on four physical qubits



with the rest of the lattice omitted. Then, at $t = t_0 + 2$, a false-negative measurement error occurred for the m -flux on star 5, resulting in the measurement of the A_s operator on star 5. No other errors occur during the error correction process from $t = 0$ to $t = t'$. This can result in the following syndromes:

Time	Star 1	Star 2	Star 3	Star 4	Star 5
t'	$m_\triangleright$	1	e	e	$m_\triangleright$
.....					
$t_0 + 4$	$m_\triangleright$	1	e	e	$m_\triangleright$
$t_0 + 3$	$m_\triangleright$	1	e	e	$m_\triangleright$
$t_0 + 2$	$m_\triangleright$	1	e	e	1
$t_0 + 1$	$m_\triangleright$	1	e	1	$m_\triangleright$
t_0	$m_\triangleright$	1	e	1	$m_\triangleright$
.....					
0	1	1	1	1	1

The anyon fluctuations on star 4 and the vacuum measured on star 5 at $t = t_0 + 2$ can be interpreted as a non-Abelian measurement error on star 5, occurring with probability q_m . While this syndrome pattern could also arise from a combination of a non-Abelian measurement error on star 5 and an e -charge pair-creation error between star 4 and an omitted star at time $t = t_0 + 2$, we disregard this possibility, as it is obviously less likely than a single non-Abelian measurement error. Alternatively, aside from a single non-Abelian measurement error, the most probable error patterns capable of producing this syndrome involve at least three errors. One such possibility consists of three m -flux pair-creation errors: one between stars 4 and 5 at time $t = t_0 + 2$, one between star 4 and an omitted star at time $t = t_0 + 2$, and one between star 5 and an omitted star at time $t = t_0 + 3$, occurring with probability p_m^3 . Another possibility involves two m -flux pair-creation errors, between star 5 and an omitted star at time $t = t_0 + 2$ and $t = t_0 + 3$, along with an e -charge pair-creation error between star 4 and an omitted star at time $t = t_0 + 2$. This occurs with probability $p_m^2 p_e$. Typically, the pair-creation error rates are comparable to or smaller than the measurement error rate, i.e., $p_e, p_m \lesssim q_m$. Therefore, errors in the quasi-stabilizer measurements of non-Abelian anyons can be reliably identified with the help of time-like heralding provided by intermediate anyon fluctuations. In contrast, a measurement error in one commuting projector does not affect the outcomes of other projectors, and thus, there will be no time-like heralding.

The identified measurement errors, along with physical errors indicated by syndrome changes at specific time steps, are placed on a 3D lattice and serve as terminals for the error correction strings. Error correction then proceeds using 3D decoders. Under pair-creation and measurement errors of non-Abelian fluxes with rates p and q , respectively, the space-like edge weights in the 3D matching decoder remain unchanged from the perfect measurement case, while time-like edges acquire weight $\frac{\ln \frac{q}{1-p}}{\ln \frac{q}{1-p}}$, following the same reasoning used in the weight assignments of Eq. C5.

2. Probability of error correction string given 3D syndromes

In principle, an optimal decoder can be constructed by considering all possible error strings E consistent with a given set of measured anyon syndromes in 3D. Similar to Eq. 4 in the main text, the probability of the homology classes can be calculated by

$$P(h|\mathbf{s}) \propto \sum_{E \in h} P(E|\mathbf{s}) = \sum_{E \in h} \frac{P(\mathbf{s}|E)P(E)}{P(\mathbf{s})} \propto \sum_{E \in h} P(\mathbf{s}|E)P(E), \quad (\text{D2})$$

where $\mathbf{s}$ denotes all terminals for the error correction strings. The probability of each string E is given by

$$P(E) = \left(\frac{p}{1-p} \right)^{|E_{space}|} \left(\frac{q}{1-q} \right)^{|E_{time}|}, \quad (\text{D3})$$

where p is the pair-creation error rate, q is the measurement error rate, and $|E_{space}|$ and $|E_{time}|$ denote the lengths of the space-like and time-like components of the error string, respectively. Obviously, $|E| = |E_{space}| + |E_{time}|$.

Accounting for multiple rounds of physical error and measurements from time t_i to t_f , and analogous to Eq. 5, $P(\mathbf{s}|E)$ can be calculated as the norm of the state

$$\prod_l [(1 - \lambda_l)(1 - A_l) + \lambda_l A_l]_{t_f} \hat{E}_{t_f} \times \cdots \times \quad (\text{D4})$$

$$\prod_k [(1 - \lambda_k)(1 - A_k) + \lambda_k A_k]_{t_i+1} \hat{E}_{t_i+1} |\psi\rangle \prod_j [(1 - \lambda_j)(1 - A_j) + \lambda_j A_j]_{t_i} \hat{E}_{t_i} |\psi\rangle \quad (\text{D5})$$

$$= \prod_{t=t_i}^{t_f} \mathcal{T} \left\{ \prod_j [(1 - \lambda_j)(1 - A_j) + \lambda_j A_j]_t \hat{E}_t \right\} |\psi\rangle, \quad (\text{D6})$$

where $|\psi\rangle$ is the state in which quantum information is encoded, $A_{j,k,l}$ are commuting projectors that define the TO as in Eq. 2, $\hat{E}_t$ is the physical error according to error correction string E at time t , and $\mathcal{T}\{\dots\}$ denotes the time-ordered product.